Facilities Planning, Design and Ergonomics

VI Semester SEE Question Papers

Course codes: IM365TDA, 18IM6C2, 16IM63, 16IM63 / 12IM71

- Aug 2025 (IM365TDA)
- Sep 2023 (18IM6C2)
- Aug 2022 (18IM6C2)
- Fast Track Oct 2020 (16IM63)
- Nov/Dec 2019 (16IM63 / 12IM71)
- Apr/May 2019 (16IM63)

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	List any two steps in making a correct location choice.	02	2	2
	1.2	Process layout is also known as _____ layout and _____ layout.	02	1	1
	1.3	List any two factors that the transportation cost is dependent on.	02	2	2
	1.4	If the centers of four departments are (10,2), (2,6), (3,2), (7,6) and has a unit cost matrix as well as flow matrix, what is the total cost of the layout.	02	2	2
	1.5	List the tools which are applicable to just in time manufacturing.	02	3	3
	1.6	Flexible manufacturing systems allows for _____ and _____ product changes.	02	1	1
	1.7	The layout where the equipment, machinery, plant and people move as necessary is known as _____.	01	2	2
	1.8	_____ technique emphasizes transportation cost matrix in the determination of facility location.	01	2	2
	1.9	A self-service cafeteria is usually positioned as _____ layout.	01	3	3
	1.10	Visibility measures used in just-in-time (JIT) would include Samples of competitor's product, including _____ and _____.	02	1	2
	1.11	_____ advocated the Systematic Layout Planning framework.	01	1	1
	1.12	List the automation systems with examples.	02	1	1

PART-B

2	a	Enumerate and analyze the characteristics of all the facilities of a supply chain to achieve excellence.	08	2	1
	b	Apply the Winning facilities planning process methodology for facilities planning of a manufacturing organization, and provide your analysis summary.	08	3	3
3	a	Explain the concept of unit load. State at least five advantages of unit load.	08	2	1
	b	Given the following dimensions of a particular type of plastic reusable containers, a trailer with inside dimensions of $240'' \times 120'' \times 120''$ is used to transport these containers. The containers are not palletized. Assume that no clearance is needed between container and the walls of the trailers. Insider dim = $20'' \times 12'' \times 12''$; Outside dim = $22'' \times 14'' \times 14''$ Each nested container = $20'' \times 12'' \times 12''$ Determine the container space utilization and storage space efficiency if the dimension of the storage opening is $24'' \times 16'' \times 14''$.	08	2	2

OR

4	a	Explain the different material handling principles that are adopted by College-Industry Council on material handling education.	08	1	1
	b	Discuss the following categories of material handling equipment with examples: i) Fixed path equipment ii) Variable path equipment	08	1	1
5		Consider the following initial layout with unit cost matrix. Use the CRAFT pair wise interchange to obtain the desirable layout. 4 2 4 4 | 1 | 5 | 4 | 4 | 2 | 3 | Initial Layout | | 1 | 2 | 3 | 4 | | A | − | 1 | 1 | 3 | | B | 2 | − | 2 | 1 | | C | 0 | 2 | − | 2 | | D | 1 | 4 | 3 | − | Flow Matrix	16	3	4

OR

6	a	Differentiate between ALDEP AND CORELAP on the basis of selection, construction and placement.	08	2	2
	b	Explain batch picking and zone picking concepts of order picking in an warehouse operations.	08	1	1
7	a	Discuss with examples and sketches, the different types of flow in a layout.	08	2	2
	b	Explain the Material handling equipment categories.	08	1	1

OR

8	a	Explain the factors for production to be considered in space planning.	08	3	4
	b	Discuss the various function in the warehouse.	08	2	3
9	a	List and explain the different efficiency indices used in evaluating the layout.	08	3	3
	b	Discuss the alternative techniques in evaluating alternative plans.	08	2	2

OR

10	a	Explain deterministic model for single facility location models.	08	1	1
	b	Emulate the impact of just in time manufacturing on facilities design.	08	3	3

18IM6C2

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations September-2023

Industrial Engineering and Management

FACILITIES PLANNING DESIGN AND ERGONOMICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	List any four objectives of Facilities Planning.	02
	1.2	List any four factors influencing plant location.	02
	1.3	List the phases of approach in Developing Alternative Material Handling System Designs.	02
	1.4	List any four elements in the definition of material handling systems.	02
	1.5	Differentiate between continuous location and discrete location problem.	02
	1.6	List the two principal approaches to order picking.	02
	1.7	List and define two main areas of Ergonomics.	02
	1.8	Differentiate between Cognitive and Physical Ergonomics.	02
	1.9	What is the difference between illuminance and luminance?	02
	1.10	List and define the two physiological responses to cold stress of the human body.	02

PART B

2	a	Explain the traditional engineering design process as applied to facilities planning.	08
	b	Elaborate the principles to be considered in designing a good plant layout.	08
3	a	Explain the classifications of various Material Handling Equipment	08
	b	List and explain the different material handling principles.	08
		OR	
4		Consider the following initial layout with unit cost matrix. Use the $CRAFT$ pairwise interchange technique to obtain the desirable layout. 6 4 4 A B 4 C D Initial Layout	

		<table><tr><th>To / From</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>--</td><td>1</td><td>1</td><td>3</td></tr><tr><td>B</td><td>2</td><td>– –</td><td>2</td><td>1</td></tr><tr><td>C</td><td>0</td><td>2</td><td>– –</td><td>2</td></tr><tr><td>D</td><td>1</td><td>4</td><td>3</td><td>– –</td></tr></table> Flow Matrix	16
5	a	Explain briefly the different type's location problems categorized on the basis of the distance between facilities.	08
	b	There are Five machines currently located in the department. They have the following coordinate locations: $P1-(1,1), P2-(6,2), P3-(2,8), P4-(3,6)$, and $P5-(8,4)$. The cost per unit distance traveled between the new machine and each existing machine is the same. The number of trips per day between the new machine and existing machines $1, 2, 3, 4$ and 5 are 10,20,25,20, and 25 respectively. Determine the optimum location for a single new facility when cost is proportional to rectilinear distance. Construct a contour line passing through the point having coordinates (3,4).	08
		OR	
6	a	Describe briefly the different types of Conventional Storage Models.	08
	b	Explain in detail the logic underlying in the simulation models for facility design.	08
7	a	Explain the three basic categories of Human-machine systems. With a Block diagram model, explain the interactions in a human-machine system.	08
	b	What are the three main components of the physical work environment? What is the difference between occupational safety and occupational health?	08
8	a	Explain briefly the factors affecting Visibility and Visual Performance.	08
	b	Provide an overview of the two general approaches that can be pursued to address the existence of a noise problem in a given facility.	08

18IM6C2

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations August-2022

Industrial Engineering and Management

FACILITIES PLANNING DESIGN AND ERGONOMICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Distinguish between the terms 'facilities location' and 'facilities design'.	02
	1.2	Facilities planning process can be improved in 3 dimensions. What are they?	02
	1.3	Write a note on life cycle cost principle of material handling.	02
	1.4	List the two elements considered in determining the size of the unit load.	02
	1.5	When a facility location problem is categorized on the basis of the distance between facilities, 4 distance measures are used. What are they?	02
	1.6	Distinguish between total cover problem and partial cover problem in the context of facilities location models.	02
	1.7	List the 3 categories of Human-Machine systems from evolutionary point of view.	02
	1.8	Mention the functional activity of the cardiac muscle and smooth muscle in human body.	02
	1.9	What is meant by color rendering? Also mention the most widely used measure for color rendering.	02
	1.10	What is meant by "startle response" in the context of auditory environment and noise?	02

PART B

2	a	Proper facilities planning along the supply chain ensures that the product will follow to the satisfaction of the end customer. In the above context, elaborate on the characteristics that the facilities in the supply chain should possess.	08
	b	A new medical facility is to be located in Bangalore. The location factors, factor rating and scores for 2 potential sites are shown in the following table. Which is the best location based on factor rating method?	

Sl.No	Location factor	Factor rating	Rating	
			L_1	L_2
1	Facility utilization	8	3	5
2	Total patient/month	5	4	3
3	Average time/ emergency trip	6	4	5
4	Land and construction costs	3	1	2
5	Employee preferences	5	5	3

08

3	a	Illustrate the container/pallet flow in conventional system, container/pallet pooling system and integrated logistics and container/pallet pooling system. Also bring out the characteristics of container/pallet pooling system.	08
	b	To simulate the development of alternatives, the ideal systems approach was proposed by Nadler. This approach consists of 4 phases. Elaborate the phases.	08
		OR	
4	a	Elucidate the procedure adopted for improvement of an existing facility using CRAFT algorithm.	08
	b	What is ALDEP in the context of layout design? Elaborate on the following in the context of the same. i. Relationship chart ii. Sweep width.	08
5	a	Identify the condition in which the location-allocation problem can be solved using quadratic assignment problem. Represent the mathematical formulation of the problem.	08
	b	How is order picking system different from unit load storage/retrieval system? Review the 2 principal approaches to order picking system.	08
		OR	
6		Write short notes on the following. i. Block stacking ii. Deep lane storage iii. Single deep storage rack iv. Double deep storage rack.	16
7	a	The capacity of human body to use energy and apply forces depends on 3 factors. What are they? Review the same.	12
	b	Distinguish between rational decision models and descriptive decision models in the context of cognitive ergonomics.	04
8	a	Evaluate the possible physiological and psychological effects of noise on human beings in industrial work environment.	08
	b	Four primary variables in “climate” strongly affect the physical well-being and comfort levels of workers. What are they? Review the possible influences of the same on human well-being in work environment.	08

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	List the constituents of 'design' of a facility.	02
	1.2	What is referred to as facilities location?	02
	1.3	Write the equation for recommended material handling system.	02
	1.4	Identify the two elements in determining the size of unit load.	02
	1.5	Distinguish between 'construction type' and 'improvement type' layout procedures.	02
	1.6	There are two strategies to follow in developing a maximally weighted planar adjacency graph. Mention them.	02
	1.7	Expand the acronym $CRAFT$.	02
	1.8	What is the first objective of warehousing?	02
	1.9	What is referred to as 'Buffer area' in the context of warehouse operations?	02
	1.10	Groover identifies four capabilities to consider the manufacturing system as flexible. List them.	02

PART-B

2	a	Describe with an appropriate illustration, the three potential dimensions for improvement of facility planning process.	08
	b	What do you understand by 'plant location'? Examine the factors to be considered in plant location decision.	08
3	a	List the ten principles adopted by College-Industry council on material handling education. Describe in detail any four of the material handling principles.	08
	b	Categorize the material handling equipment broadly.	08
		OR	
4	a	Write a brief note on the following: i) Stackability ii) Nestability.	08
	b	Illustrate container/pallet flows in following systems: i) Conventional system ii) Pooling system iii) Integrated contract logistics and container/pallet pooling.	08

5	a	Write a note on the following: i) Procedural differences between $CORELAP$ and $ALDEP$. ii) Sweep width in the context of $ALDEP$. iii) Listing warehousing opportunities in a logistics network.	09
	b	Illustrate through a figure, different typical warehouse functions and flows.	07
		OR	
6	a	Analyze the steps required to determine the total space requirements for receiving and shipping areas.	08
	b	Write note on the following in the context of order picking: i) Discrete picking ii) Batch picking iii) Zone picking iv) Wave picking.	08
7	a	Choose ten main components of a kitchen (ex: oven, sink). Develop a relationship chart of the ten components. Sketch a kitchen containing the ten components and arrange them based upon relationship chart findings.	12
	b	List at least six factors for consideration in planning material flow.	04
8	a	Illustrate the following alternative layouts for a flexible manufacturing system: i) FMS with centralized layout ii) FMS based on robot handling system.	06
	b	Explain the five rules of thumb used in designing inprocess handling and storage systems.	10

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An autonomous Institution affiliated to VTU)

VI / VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Facilities planning are subdivided into the subjects of facilities _____ and facilities _____.	02
	1.2	Manufacturing of a standard product in large quantity is preferred in _____ layout.	02
	1.3	Principle of 'Unit load' states that _____ should be moved in _____.	02
	1.4	List the objectives of material handling.	02
	1.5	The appropriate goal of material handling is to design the most efficient _____ handling system at the _____ cost.	02
	1.6	List the inputs required for $CRAFT$ (Computerized Relative Allocation of Facilities Technique).	02
	1.7	Just in time was successfully implemented by _____ and invented by _____.	02
	1.8	Transfer lines are often used for _____ volume and highly _____ manufacturing environment.	02
	1.9	The effectiveness of the _____ layout and _____ handling influences the product mix and design.	02
	1.10	List the components of flexible manufacturing systems.	02

PART B

2	a	Analyze the following characteristics of the facilities planning in the context of a new manufacturing firm: i) Flexibility ii) Modularity iii) Upgradability iv) Adaptability.	08
	b	A manufacturer of a farm equipment is considering 3 locations A, B and C for a new plant. Cost studies show that the fixed costs per year at the sites are $Rs.2{,}40{,}000/-$, $Rs.2{,}70{,}000/-$ and $Rs.2{,}52{,}000/-$ respectively. The variable costs per year Rs. 100 per unit, $Rs.90$ per unit and $Rs.95$ per unit respectively. If the plant is designed to have an effective system capacity of 2500 units, find the most economic location.	08

3	a	Define material handling? List the objectives of material handling.	08
	b	State the guidelines in selecting material handling equipments?	08
		OR	
4	a	State and explain different principles of plant layout.	10
	b	Discuss the concept of unit load with an example.	06
5		An Auto Parts warehouse has requested that a new layout be designed for their main warehouse. The ware house has 8 major activity centers. The current building area can accommodate 10×24 unit squares. Develop the new layout using *ALDEP* using the data given below.	

Code	1	2	3	4	5	6	7	8
Area (m^2)	12800	8800	9200	12000	8800	13200	13200	8000

Storage	---						
Counter	*X*	---					
Paint Room	*O*	*O*	---				
Receiving	*A*	*X*	*U*	---			
Office	*U*	*E*	*U*	*U*	---		
Parts Bins	*O*	*A*	*O*	*U*	*U*	---	
Tailpipe Racks	*O*	*I*	*I*	*U*	*U*	*I*	---
Restroom	*O*	*O*	*O*	*U*	*O*	*O*	*O*	---

Closeness	*A*	*E*	*I*	*O*	*U*	*X*
Value	64	16	4	1	0	−1024

Minimum Department Preference (MDP) = I = 4; Sweep Width = 2; Number of iterations to be performed, $N = 1$; Assume Scale: 1 square = $400m^2$; Length × Width for the layout: 10×24. 16

OR

6	a	Explain functions of warehouse operations in the following areas: i) Receiving ii) Inspection and quality control.	08
	b	Discuss with example any two principles of order picking in warehouse operations.	08
7		A certain shop manufactures a product using different machines. The details of the machines are given below.	

Machine No.	*Machine Area* (m^2)	*No. of Machines*	*Machine Code*	*Production rate/hour*
1	4×3	1	*O*1	60
2	4.25×3	2	*O*5, *O*6	70
3	8.95×3	1	*O*8	80
4	7×3.5	1	*O*9	65
5	3.5×2.4	1	*O*11	50
6	4.6×3	2	*O*13, *O*14	60
7	11×4	1	*O*15	55
8	2.36×2	1	*O*17	45
9	8.43×3.57	1	*O*21	50
10	3.3×3	1	*O*23	45
11	4.48×2.9	1	*O*26	50
12	3.3×2	3	*O*28, *O*29, *O*30	60
13	6×4.15	1	*O*31	70

		The order of flow is $O8-O23-O15-O11-O14-O29-O17-O1-O6-O13-O28-O26-O31-O5-O21-O9-O30$ Maximum dimensions of a product = $30mm \times 500mm$; No. of hours per shift = 8; No. of shifts per day = 3; Category of material handling = Light; WIP details = 3 high stacking at each WIP area; Services to be included = Maintenance, Tool crib, Inspection, Supervision and Amenities. Draw an appropriate shop layout to a suitable scale and position machines for straight line flow within it, providing administrative area involving 5 executives and 12 supporting staff, a conference hall and a generator room.	16
8	a	Explain with an example fixed automation systems for a dial indexing machine.	08
	b	Analyze the applications for which flexible manufacturing systems is applied.	08

16IM63

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	List the three potential dimensions for improvement of facility planning process.	02
	1.2	Distinguish between the terms "facilities location" and "facilities design".	02
	1.3	Mention the material handling system equation that helps in guiding the development of alternative material handling system designs.	02
	1.4	Identify the two elements in determining the size of the unit load.	02
	1.5	Distinguish between the features of layout for a product family department and layout for a process family department.	02
	1.6	What is meant by the term "sweep width" in solving a layout issue using *ALDEP* algorithm?	02
	1.7	Identify the equipment used as an interface between a dock at a given height and variable height carriers.	01
	1.8	What is meant by 'Honeycombing' in the context of warehouse operations?	02
	1.9	List the external factors that affect the facility planning process.	02
	1.10	List the basic components of a flexible manufacturing system.	02
	1.11	Expand *CRAFT*.	01

PART B

2	a	Evaluate the impact of facilities planning on the competitiveness of manufacturing facilities.	08
	b	Analyze the major factors that influence a typical plant layout.	08
3	a	Summarize the steps involved in the process of designing material handling systems.	08
	b	Illustrate and explain different stacking patterns for different pallet sizes.	08
		OR	
4	a	Categorize the material handling equipment based on the functions performed.	08

	b	For each of the following material handling principles, explain what it means and give a specific example of how it can be applied: i) Ergonomic principle. ii) Unit load principle. iii) Work principle. iv) Space utilization principle.	08
5	a	Bring out the basic differences between construction type and improvement type algorithms.	08
	b	Provide a review of the following: i) $COFAD$ ii) $PLANET$.	08
		OR	
6	a	Analyze the principles of 'order picking' operation.	08
	b	Evaluate the principles to be considered in accomplishing storage area objectives.	08
7		Consider a food restaurant you visit often. Describe in detail the arrangement of aisles for customers (the service line, the booth area, etc) and the arrangement of aisles for the workers (area behind the counter) by sketching a layout. Draw the flow of the customers and employees on the layout. Explain how the flow of the workers and the customers would change if the layout was different. Is there adequate space for the customers and employees to operate efficiently? Can any improvement in the layout create a more efficient flow? Describe any visual management approaches being used.	16
8	a	Analyze the impact of Just-in-time philosophy on the design of facility layout and material handling systems.	08
	b	Describe the single stage multimachine systems in terms of the resources involved.	08

Service Operations Management

VI Semester SEE Question Papers

Course codes: IM365TDB, 21IM64D2, 16IM6C2

- Aug 2025 (IM365TDB)
- Sep/Oct 2024 (21IM64D2)
- Grade Improvement Nov/Dec 2020 (16IM6C2)
- Apr/May 2019 (16IM6C2)

IM365TDB

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Industrial Engineering and Management

SERVICE OPERATIONS MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Define service blueprinting.	02	1	1
	1.2	List any two examples of services with high labor intensity and high interaction.	02	1	2
	1.3	What do you mean by a “moment of truth” in service delivery?	02	2	3
	1.4	List any four traits of excellent service personnel.	02	1	1
	1.5	What role does empathy play in customer service experience?	02	2	1
	1.6	What are the implications of poor capacity planning in service systems?	02	2	2
	1.7	Distinguish between front-stage and back-stage service activities.	02	2	3
	1.8	List two major benefits of technology in service design.	02	1	2
	1.9	Name two Industries where SERVQUAL can be effectively used.	02	2	1
	1.10	Mention any two types of customer expectations.	02	2	4

PART-B

2	a	Use hotel industry as an example, how do you differentiate Capability as opposed to Commodity. Give insights on Process styles, Service offer, Major challenges faced.	10	3	1
	b	Identify any 3 types of services and enumerate the challenges faced by these services?	06	1	1
3	a	You are hired as a consultant for improving the operations of a public transport system. Identify key tactical and strategic challenges you may encounter. Recommend a roadmap for improvement.	12	5	3
	b	Describe how “Service as a concept” can be used as a strategic tool.	04	2	1
		OR			
4	a	Explain the following customer types using a quadrant model: The Hostage, The Ally, The Anarchist, The Champion. Suggest one strategy to handle each.	10	3	3
	b	Stakeholder management is an important aspect of any service operations. Having many customers as allies can be the best thing that can happen to any service/business. Recommend how allies can be created among various customer types.	06	3	3
5	a	Illustrate with an example the downsides of expectation – perception approach to customer perceived quality.	04	3	2

	b	Statement 1: Satisfied customers are a sign of success and an impetus for productivity. Statement 2: Satisfied customer is someone who has availed the services provided. Considering the previous statements, explain the concept of "Service Confidence and how it can be enhanced" taking example of services offered by police.	06	4	2
	c	Identified any two internal causes for customer dissatisfaction and also represent following using appropriate block diagrams. i) The satisfaction continuum expressed in _5 to _5 scale. ii) Perception-expectation simplified gap model.	06	4	1
		OR			
6	a	Using a block diagram, explain the aspects of service quality using Airline Industry as an example with specific emphasis on any two of the below mentioned topics. i) Inputs ii) The service process iii) The service experience iv) Outcomes	08	3	3
	b	Assume a customer touch point example of your choice and briefly explain it. Defend your understanding of how customer satisfaction can be enhanced/maintained (Customer touch point considered) for the below mentioned use cases using _5 to _5 scale template. i) Adequate performance satisfies the customer ii) Using enhancers to delight the customer iii) A dissatisfied outcome iv) Enhancers compensate for failures	08	4	2
7	a	With respect to service product variety, discuss the following terms i) Runners ii) Repeaters iii) Strangers Also, comment on how organizations handle these runners, repeaters and strangers.	08	3	3
	b	Discuss how service processes can be repositioned.	08	3	3
		OR			
8	a	Service capacity is defined as the maximum level of value-added activity over a period of time that a service process can consistently achieve under normal operating conditions. Organization would keep experimenting with terms "Consistently" and "Under normal operating conditions". Comment from your life experience how these affect the Quality.	08	3	2
	b	Ensuring customer satisfaction enhances the performance of the organization. Customer satisfaction can be measured using several methods. Explain four such methods as per the below mentioned format.			

Sl. No.	Method Name	Explanation	Useful for organizations such as
1			
2			
3			
4			

			08	3	3

9	a	Depict a service and its challenges. Explain how you wish to use SERVQUAL and MoSCoW prioritization matrix to assess the requirements and measure it. Justify its usage to a Naïve Service Operations Manager running a typical service that you have chosen.	10	5	4
	b	How might a hotel chain's service operations manager use various performance measure to enhance overall performance?	06	3	3
		OR			
10	a	Difference between level and chase strategy with appropriate examples.	08	3	2
	b	What are the types of targets used in performance management, and how do they differ from each other?	08	4	3

21IM64D2

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations Sept/Oct – 2024

Industrial Engineering and Management

SERVICE OPERATIONS MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	List two metrics used to judge the success of a service operation.	02	1	1
	1.2	Why are standardizing service processes often difficult in service operations management?	02	2	2
	1.3	How can customers be classified in service operations management?	02	2	3
	1.4	Why is customer retention important in service operations?	02	2	1
	1.5	What factors influence our confidence in a service before we receive it?	02	2	2
	1.6	Highlight the two key benefits of e-procurement over traditional procurement.	02	2	4
	1.7	What is Process mapping?	02	2	5
	1.8	Distinguish between commodity and capability processes.	02	2	1
	1.9	Compare a restaurant chain which has a centralized kitchen and many outlets across Bengaluru as opposed to a restaurant which has a decentralized kitchen in terms of advantages.	02	4	2
	1.10	List different types of Benchmarking.	02	2	3

PART-B

2	a	Explain the 'service concept' using examples from popular Indian food brands such as Haldiram's and MTR. How do these brands' service approaches illustrate the key elements of an effective service concept in the Indian market?	10	3	2
	b	Considering yourself as a Service Operations Manager for an online retail store, highlight on the challenges that you might encounter.	06	2	2
3	a	Illustrate the impact of a well-defined service concept on customer experience using examples from different industries. How does a clear service concept contribute to achieving competitive advantage?	08	3	1
	b	Evaluate the role of managing customer relationships in service operations in India. What practices can organizations adopt to effectively manage and improve these relationships?	08	4	1
		OR			
4	a	Identify and describe the different categories of customers based on their behavior or attitude. How do these classifications help businesses tailor their strategies?	10	2	1

	b	Explain how managers can develop good customer relationships in a $B2C$ service environment.	06	2	2
5	a	Discuss the factors contributing to gaps between customer expectations and perceptions in service delivery. What are the implications of these gaps on perceived service quality?	08	2	1
	b	What are service quality factors? Briefly describe any six service quality factor with example	08	2	4
		OR			
6	a	What strategies can be employed to effectively manage service supply chains?	08	2	3
	b	Outline the primary challenges that service operations managers face in managing complex and often messy supply chains, particularly when dealing with intermediaries. How can these challenges be effectively addressed?	08	2	2
7	a	Imagine a potential customer navigating the website of a fashion retail outlet to select and order a dress. How can understanding the nature of service processes help in designing a better customer experience? Provide specific examples of how this understanding can enhance the online shopping journey.	08	3	3
	b	With respect to service product variety, discuss the following terms i) Runners ii) Repeaters iii) Strangers Also, Comment on how organizations handle these runners, repeaters and strangers.	08	2	1
		OR			
8	a	For a service with significant front-office customer interaction, illustrate strategies to effectively manage bottlenecks and queues. How can these strategies improve the customer experience?	08	3	4
	b	Discuss how service processes can be repositioned.	08	2	1
9	a	Choose a specific service and describe its associated challenges. Explain how you would use $SERVQUAL$ prioritization matrix to assess and measure its requirements. Justify use of the tool to a novice service operations manager in the context of the chosen service.	06	3	5
		How can a hotel chain's service operations manager use various performance measures to improve overall operational performance? Provide examples of performance metrics and their impact.	10	2	4
		OR			
10	a	Differentiate between the level strategy and the chase strategy in capacity management, providing relevant examples for each approach.	06	4	1
	b	What are the different types of targets used in performance management, and how do they differ from one another? Explain the significance of each type in managing performance effectively.	10	2	1

16IM6C2

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Grade Improvement Examinations Nov/Dec-2020

Industrial Engineering and Management

SERVICES OPERATIONS MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Identify two significant responsibilities of service operations manager.	02
	1.2	Comment on the consequences if a budget hotel follows capability process as per capability commodity model.	02
	1.3	Imagine you are going to watch an *IPL* cricket match at Namma Bengaluru's very own Chinnaswamy Cricket stadium, list 4 aspects that you associate regarding service experience.	02
	1.4	When distinguishing airlines based on the type of operations, Low cost airlines are typically associated with _____, _____, _____ & _____.	02
	1.5	Comment on challenges of running an *NGO*.	02
	1.6	Assume you are being interviewed for your short but successful career in the field of "Service operations management". One of the questions from a young enthusiast is "How to develop a culture of service quality in an organization?" Comment on at least 4 aspects.	02
	1.7	From your experience, give two examples where technology is used to enhance customer experience.	02
	1.8	List 2 differences between service operations and manufacturing operations organizations.	02
	1.9	Comment on the tools that you use to enhance customer satisfaction.	02
	1.10	Online shopping is the new trend among most of youngsters. Identify the 4 moments of truth for a product or service bought via online shopping as listed below. Zero Moment of truth: First Moment of truth: Second Moment of truth: Third Moment of truth:	02

PART B

2	a	Use hotel industry as an example, how you differentiate capability as opposed to commodity. Give insights on process styles, service offer, Major challenges faced.	10
	b	Identify any 3 types of services and enumerate the challenges faced by these services?	06

3	a	Assume yourself in a position of a service operations manager in a luxury resort in Maldives. Evaluate the challenges in managing the operations both tactically as well as strategically.	10
	b	Describe how "Service as a concept" can be used as a strategic tool.	06
		OR	
4	a	Explain how managers can develop Good customer relationship in a B2C service environment.	04
	b	Explain different types of customer types you might encounter in a service environment using appropriate block diagram.	06
	c	Stakeholder management is an important aspect of any service operations. Having many customers as allies can be the best thing that can happen to any service / business. Recommend how allies can be created among various customer types.	06
5	a	Illustrate with an example the downsides of expectation perception approach to customer perceived quality.	04
	b	Statement 1: Satisfied customers are a sign of success and an impetus for productivity Statement 2: Satisfied customer is someone who has availed the services provided. Considering the previous statements, Explain the concept of "Service Confidence and how it can be enhanced" taking example of services offered by police.	06
	c	Identify two internal causes for customer dissatisfaction and also represent following using appropriate block diagrams. i) The satisfaction continuum expressed in_5 to_5 scale. ii) Perception-Expectaion simplified gap model.	06
		OR	
6	a	Using a block diagram , Explain the aspects of service quality using Airline Industry as an example with specific emphasis on any two of the below mentioned topics i) Inputs ii) The Service process iii) The service Experience iv) Outcomes	08
	b	Assume a customer touch point example of your choice and briefly explain it. Defend your understanding of how customer satisfaction can be enhanced/maintained (Customer touch point considered) for the below mentioned use cases using _5 to_5 scale template. i) Adequate performance satisfies the customer ii) Using enhancers to delight the customer iii) A dissatisfied outcome iv) Enhancers compensate for failures v) Dissatisfaction shifts the zone of tolerance vi) Delight also shifts the zone of tolerance.	08

7	a	With respect to service product variety, discuss the following terms i) Runners ii) Repeaters iii) Strangers Also, Comment on how organizations handle these runners, repeaters and strangers.	08
	b	Discuss how service processes can be repositioned.	08
8	a	Service capacity is defined as the maximum level of value added activity over a period of time that a service process can consistently achieve under normal operating conditions. Organization would keep experimenting with terms "Consistently" and "Under normal Operating Conditions". Comment from your life experience how these affect the quality.	08
	b	Ensuring customer satisfaction enhances the performance of the organization. Customer satisfaction can be measured using several methods. Explain four such methods as per the below mentioned format.	08

Sl No	*Method Name*	*Explanation*	*Useful for Organizations Such as*
1			
2			
3			
4			

16IM6C2

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

SERVICES OPERATIONS MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	With 2019 loksabha assembly elections around the corner, its our responsibility to vote for the right candidate who can serve better. Comment on any two challenges of providing public services ($G2C$).	02
	1.2	Distinguish between service operations with regard to a low-cost airline and a Full-service airline [any two differences] .	02
	1.3	Report the major challenges in providing service with regard to $B2B$ and $B2C$.	02
	1.4	Give any two implications of technological advances in educational sector.	02
	1.5	Expectations of a service are dynamic in nature and are influenced by _____ and _____.	02
	1.6	From your experience, give two examples where technology is used to enhance customer experience.	02
	1.7	Write any two points highlighting the importance of service process design.	02
	1.8	Assuming your family is running the food court at your college campus, where the challenges are different in comparison to a usual restaurant business in any business district. List the rewards and targets for your employees in such a scenario?	02
	1.9	Identify any two operational performance measures for smooth functioning of a typical bank.	02
	1.10	Identify any one reason why bench marking practice is criticized as industrial tourism.	01
	1.11	Identify any two types of targets any service firm should work towards.	01

PART B

2	a	Describe the six tactical challenges faced by service operations managers in a typical service environment.	08
	b	Your friend wants to be an entrepreneur and wants to provide a particular service in a tier *II* city like Mysore. He/she needs your suggestion to enhance the service experience. Explain the aspects of service experience that you would emphasize on in order to suggest your friend. Defend whether your suggestion would change to some extent if the same service were to be provided in a tier *I* city like Bengaluru.	08

3	a	A service concept is shared understanding of the nature of the service provided and received. Using an example of a theme park, explain the following: i) Organizing the idea. ii) The service experience iii) The service outcome iv) The service operation v) The value of the service.	10
	b	Describe how "Service as a concept" can be used as a strategic tool.	06
		OR	
4	a	Illustrate with an example how important is retaining good customer for the success of any business.	08
	b	Write short notes on(Use quadrants to showcase your understanding): i) Classification of customer types ii) Creating allies among the various customer types.	08
5	a	Sketch the touch points in the customer's journey of the following case. "A patient with an appointment to see a doctor for a routine medical examination".	05
	b	In consideration for the above case, illustrate how customer satisfaction can be enhanced at various touch points in the customer journey.	06
	c	Taking an example of a bank, recognize the importance of Hygiene factors, Critical factors, Neutral factors and Enhancing factors.	05
		OR	
6	a	Discuss the role of innovation and internet technology in developing the customer experience.	08
	b	One of the key decisions for procurement function is choosing/selecting a supply partner and particularly for those who handle critical elements of supply chain. Discuss the eight criterion for supplier selection excluding cost and quality.	08
7	a	Showcase your understanding of Runners, Repeaters and Strangers for providing following service in the given format below. Also, Comment on which among these (Runners, Repeaters and Strangers)is least efficient and difficult to handle and the reasons for it.	10
	b	Describe how can service processes be repositioned?	06
8	a	Write short notes on(Minimum four points): i) General rules for managing bottlenecks ii) Principles of queue management (perceived waiting time vs actual waiting time).	08
	b	Evaluate the need for "Metric Benchmarking" processes and the shortfalls while doing it.	08

Service	*Runner*	*Repeaters*	*Strangers*
Car service			
International Airline			
Hospital			

Global Supply Chain Management

VI Semester SEE Question Papers

Course codes: 21IM62

- Sep/Oct 2024 (21IM62)

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Examinations Sept/Oct– 2024

Industrial Engineering and Management

GLOBAL SUPPLY CHAIN MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables permitted.

PART-A

			M	BT	CO
1	1.1	The most limited scope over which the strategic fit is considered is one operation within a functional area within a company. This is referred to as _____ scope	01	2	1
	1.2	Draw the cost-responsiveness efficient frontier curve	02	2	1
	1.3	_____ cycle occurs at the manufacturer / supplier interface and includes all processes to ensure that materials are available for manufacturing to occur according to schedule	01	2	1
	1.4	The coordinate location of a manufacturer is $(40, 60)$. The coordinate location of its customer is $(20, 10)$. The manufacturer is required to supply 200 compressors to this customer. The cost of transporting one compressor is *Rs* 100. Using the gravity location model, determine the distance between the manufacturer and supplier, and the total transportation cost.	02	3	2
	1.5	Graphically depict variation of transportation cost and response time with number of facilities	02	2	2
	1.6	Enumerate the risks that are to be considered during network design.	02	2	2
	1.7	If the manager at B&M manages inventory such that the store does not run out of inventory in 8 out of 10 replenishment cycles, then stores achieve a CSL of _____%.	01	3	3
	1.8	What are the two factors which determine appropriate level of safety inventory?	02	2	3
	1.9	_____ contract allows the buyer to modify the order (within limits)as demand visibility increases closer to the point of sale	01	2	3
	1.10	Mention any two benefits of effective sourcing decisions	02	2	3
	1.11	What is continuous replenishment in supply chain management?	02	2	3
	1.12	Identify one example of a collaborative planning activity in CPFR	02	2	4

PART-B

2	a	Write the block diagram to depict the framework for structuring the drivers of supply chain performance. Choose an Indian organization and discuss the structuring of its drivers.	08	3	1
	b	Describe the cycle and push/pull views of a supply chain. Which of the two views is very useful when considering strategic and operational decisions related to Supply chain design	08	2	1

3	a	Discuss in detail about the role, factors influencing with respect to facility location for a pharmaceutical Supply chain.	08	3	2
	b	Illustrate and discuss three performance characteristics related to service for the following distribution network design options, using diagrams: (i) Manufacturer storage with direct shipping (ii) Distributor storage with last-mile delivery	08	2	2
		OR			
4	a	Identify and describe the factors influencing Network Design decisions	08	2	2
	b	Discuss how you evaluate the performance of a distribution network along the two dimensions namely: customer needs and the cost.	08	2	2
5	a	Weekly demand for Legos at Toys store is normally distributed with a mean of 2500 boxes and a standard deviation of 500. The replenishment lead time is two weeks. i. Assuming a continuous review replenishment policy, evaluate the safety inventory that the store should carry to achieve a CSL of 90% ii. If the store manager decides to follow a periodic review policy to manage inventory of Legos and plans to review inventory after every 4 weeks, for a CSL of 90%, how much of safety inventory should the store carry? What should its Order upto level (OUL) be? iii. Compare the above 2 replenishment policies and comment on requirement of safety inventory	10	3	3
	b	Weekly demand for 12" frames at the Frame Shop is normally distributed with a mean of 250 and a standard deviation of 150. The store manager continuously monitors inventory and currently orders 1,000 frames each time the inventory drops to 600 frames. The manufacturer currently takes two weeks to fill an order. How much safety inventory does the store carry?	06	3	3
		OR			
6	a	Explain the role of cycle inventory and safety inventory on supply chain with suitable examples.	10	3	3
	b	Weekly demand for microwave ovens at Best Buy store is normally distributed with mean of 2500 and standard deviation of 500. It takes two weeks to supply a Best Buy order. Best Buy targeting a CSL of 90 percent and monitors its inventory continuously. How much safety inventory of microwave ovens should Best Buy carry?	06	3	3
7	a	Analyze the factors affecting transportation decisions with the aid of a suitable example	10	2	3
	b	What are the reasons for outsourcing? Illustrate with an example	06	2	3
		OR			
8	a	Identify and discuss the factors considered when scoring and assessing suppliers in a supply chain context.	10	2	3
	b	Using examples, describe revenue-sharing and quantity flexibility contracts.	06	3	3

9	a	Describe supply chain co-ordination and the bull whip effect and their impact on supply chain performance	10	2	4
	b	How does Customer Relationship Management (CRM) contribute to supply chain efficiency?	06	2	4
		OR			
10	a	What is vendor managed inventory (VMI)? How it is different from continuous replenishment strategy? How is the manufacturer benefitted out of VMI?	10	2	4
	b	What does CPFR stand for in supply chain management? How does CPFR improve supply chain efficiency?	06	2	4

Lean Manufacturing Systems

VI Semester SEE Question Papers

Course codes: 21IM65E1, 12IM6D2

- Sep/Oct 2024 (21IM65E1)
- Apr/May 2018 (12IM6D2)
- Apr/May 2017 (12IM6D2)
- Apr/May 2016 (12IM6D2)

21IM65E1

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Examinations Sept/Oct – 2024

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Define Lean Manufacturing.	02	2	1
	1.2	Mention the two pillars of the Toyota Production System (*TPS*)?	02	2	1
	1.3	Why is inventory needed in a manufacturing system?	02	2	1
	1.4	What is a Kanban system?	02	1	1
	1.5	List two benefits of lead-time reductions.	02	2	2
	1.6	What are the three fundamental issues of cultural change in lean implementation?	02	2	2
	1.7	What is the purpose of conducting a system-wide evaluation of the present state in lean implementation?	02	2	2
	1.8	What is Kaizen and its role in lean manufacturing?	02	2	2
	1.9	What is Hoshin-Kanri Planning?	02	1	1
	1.10	Why is it important to sustain the gains achieved through lean implementation?	02	2	2

PART-B

2	a	Explain the concept of Lean Manufacturing and discuss its primary goals.	08	2	1
	b	Discuss Taiichi Ohno's thoughts on the Toyota Production System and its significance in modern manufacturing.	08	2	1
3	a	Discuss the need for inventory in manufacturing and the disadvantages associated with high inventory levels.	08	3	4
	b	Explain the concept of Kanban and describe the process of Kanban calculations in a lean manufacturing system.	08	3	2
		OR			
4	a	Explain the need for inventory in a manufacturing system and discuss the various types of inventory that are typically maintained.	08	3	4
	b	How do finished goods inventory calculations help in lean manufacturing, and what factors are considered in these calculations?	08	3	2
5	a	Analyze the benefits of lead-time reduction in manufacturing and describe techniques used to achieve these reductions.	08	3	2
	b	Discuss the cultural change fundamentals necessary for implementing lean manufacturing and the challenges associated with these changes.	08	2	3

		OR			
6	a	Trace the historical evolution of lead time in manufacturing and discuss its changing significance over time.	08	3	3
	b	Examine some cultural aspects that are essential for a successful lean implementation.	08	3	3
7	a	Outline the steps involved in implementing a lean project and discuss the importance of each step in the process.	08	2	3
	b	"After implementing lean practices, an organization has seen significant improvements. However, there is concern about sustaining these gains over time. Discuss the importance of sustaining the gains in a Lean environment. How would you address the existence of process gains and losses to ensure long-term success?"	08	4	4
		OR			
8	a	"A manufacturing company is struggling with inefficiencies in its production line. As a Lean consultant, you are asked to guide the company through the Lean implementation process. Discuss how you would assess the three fundamental issues related to cultural change in the company, and outline the steps you would take to complete a system-wide evaluation of the present state."	08	4	4
	b	Explain the concept of Kaizen and its role in the continuous improvement process within lean manufacturing.	08	2	3
9	a	"A manufacturing company is transitioning to Lean 4.0. Explain how the traditional dimensions of lean manufacturing, such as waste reduction, just-in-time production, and continuous improvement, are influenced by the technologies and principles of Industry 4.0. Provide specific examples of how these dimensions can be enhanced by digital transformation."	08	4	3
	b	Describe Hoshin-Kanri Planning and its importance in aligning organizational goals with lean initiatives.	08	2	3
		OR			
10	a	"During the implementation of *Lean* 4.0, a manufacturing company faces several challenges, including resistance to change, data management issues, and high implementation costs. Discuss these challenges in detail and propose practical solutions to overcome them."	08	3	3
	b	Discus the importance of sustaining the gains achieved through lean implementation and it strategies to prevent process loss.	08	2	3

12IM6D2

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART A

1	1.1	At the heart of *JIT* is a deep understanding and control of _____.	01
	1.2	The makeup of the typical production cell line and how _____ is handled is a key manufacturing aspect of *TPS*.	01
	1.3	Define buffer stock.	01
	1.4	State Little's law.	01
	1.5	As per *KP* methodology, what is a problem?	01
	1.6	Name one of the three requisite skills that are required for the leader of lean initiative.	01
	1.7	Cite a reason to implement lean using one value stream at a time approach.	01
	1.8	Future state value stream map will _____ production internally.	01
	1.9	The key uniqueness in the Hoshinkanri planning is the concept of _____.	01
	1.10	The two strongest techniques to sustain the gains once they are found are through _____ and _____ simplification.	01
	1.11	Write two differences between *KANBAN* and *CONWIP*.	02
	1.12	In a manufacturing organization the production time is 23.3 hours, planning time in 24 hours delivery time is 2 hours and the typical queue of 16 hours. Assuming $\alpha = 0.05$ calculate cycle stock. takt time is 1 minute.	02
	1.13	What are the approaches to reduce first piece lead time and total lead time?	02
	1.14	_____ and _____ are the two extremely powerful detractors that will rear their ugly hands time and again as an organization pursues the lean journey.	02
	1.15	Name any two characteristics of 'Goals' that are created during lean implementation.	02

PART B

2	a	Write a detailed note on pull production technology.	08
	b	Discuss the following revolutionary concept used in the *TPS*: The supply of value to the customer.	08
		OR	
3	a	The *TPS* is not a complete manufacturing system. Critically analyze the statement.	08
	b	Prepare a comparative analysis between lean and the Toyota production system.	08

4	a	"To reduce inventory levels reduce variation". Justify the statement.	06
	b	Enumerate the six rules of *KANBAN* and the several unique functions that are provided.	10
		OR	
5	a	Summarize the people aspects of the basic needs to execute the *TPS* (quality control).	09
	b	Elaborate the concept of Jidhoka as a pillar of quantity control. List the techniques under Jidhoka pillar.	07
6	a	List and explain briefly the benefits of lead time reductions. Illustrate with examples.	08
	b	Bring out the importance of presence of talented problem solvers as a fundamental issue of cultural change to be addressed during lean implementation.	08
		OR	
7	a	The interdependence of activities is a unique aspect of lean implementation initiative. As a team member of lean implementation group how do you handle this issue?	04
	b	Select a process of your choice with minimum three stages. Develop the present state value stream map with usual notations and symbols. Make suitable assumptions about the data.	12
8		For a small scale manufacturing organization of about 100 employees how do you perform the assessment of the three fundamental issues to cultural change? Prepare a report to be submitted to the management for lean implementation.	16
		OR	
9		Assume that you are chosen as the sensei. How do you evaluate a manufacturing company based on the five precursors to implementing a lean initiative? Prepare a report of evaluation which becomes a significant part of the implementation plan.	16
10	a	Why are goals and goal deployment so important? Discuss.	08
	b	Illustrate with a sketch the Hoshinkanri planning model Enumerate unique strengths of Hoshinkanri planning.	08
		OR	
11	a	"Transparency is the concept that lets you see in real time right now what is happening in your process". Elaborate with examples.	08
	b	Explain the various means of achieving process gains by either reducing the variation in the product, the process or both.	08

12IM6D2

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

7 Answer FIVE full questions from Part B.

PART A

1	1.1	The two pillars of TPS are _____ and _____.	02
	1.2	Lean is also about _____ and _____.	02
	1.3	As per Little's law lead time is the ratio of _____ and _____.	02
	1.4	Total number of finished goods Kanban is the sum of (_____) + (_____) + (_____), divided by _____.	02
	1.5	Expand $SMED$ and OEE.	02
	1.6	Takt is the ratio of _____ and _____.	02
	1.7	List any four out of seven wastes.	02
	1.8	Define a push system.	02
	1.9	The three fundamental issues to be assessed to cultural changes are i) Do we have the _____ to make this a success? ii) Do we have the _____ to make this a success? iii) Do we have the _____ to make this a success?	02
	1.10	Muda,, Mura and Muri means _____, _____ and _____ respectively.	02

PART B

2	a	Define Lean.	02
	b	Discuss the ohno's thought about the Toyota production system.	06
	c	Explain in detail where lean will not work?	08
		OR	
3	a	Explain clearly the two pillars of the TPS.	08
	b	Write a note on: i) TOC ii) Crosby's approach to zero defects.	08
4	a	An insurance company processes 9600 claims per year. The company will not work for 4 weeks/year. Calculate the WIP if the average processing time of a claim is 21 days.	06
	b	What is so bad about Inventory? In this context, discuss the two basic questions which are frequently overlooked by the managers.	10
		OR	

5	a	Discuss the following: i) $5S$ techniques ii) MSA.	08
	b	What is TPM? Explain the five pillars of TPM.	08
6	a	Illustrate the role played by $SMED$.	04
	b	Describe the seven techniques to reduce lead time.	12
		OR	
7	a	With a sketch explain Jung's scales for personality types.	07
	b	Discuss the three basic issues of cultural change.	09
8		Write a note on the following i) Seven wastes ii) VSM iii) Spaghetti diagram.	06 05 05
		OR	
9	a	List all the steps used as a prescription for the lean implementation and explain any one of them.	10
	b	With a sketch explain Kaizen spiral.	06
10		Write a note on the following: a) Process gain b) Poka – yoke c) Sustaining the gains d) Goal characteristics.	16
		OR	
11	a	Explain Hoshin – Kawi planning model.	08
	b	Analyze the concept "how do we know there is a loss?"	08

12IM6D2

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	What is the popular definition of lean?	02
	1.2	What did Ohno say about the Toyota Production system?	02
	1.3	To reduce inventory levels, reduce_____.	01
	1.4	List the three forms of buffers.	02
	1.5	The foundation of high quality has two strategies. First is the _____ of the workforce.	01
	1.6	Two key characteristics that few businesses measure in any form but that all businesses want are flexibility and_____.	01
	1.7	Lead time as a metric was not discussed much at all until the book _____ (Lean Enterprise Institute, 1998) heightened peoples awareness of this powerful concept.	01
	1.8	The _____ is a simple yet powerful tool to visualize movement and transportation in a shop floor.	01
	1.9	On the balancing chart, one can quickly see three things: how much time is wasted _____ and the bottleneck.	01
	1.10	The leader and the_____ are far and away the two most important people in the lean implementation effort.	01
	1.11	As part of the complete a system wide evaluation of the present state, the leadership makes_____ evaluation of the entire manufacturing system.	01
	1.12	Ohno says that_____ is the basic condition. It is the foundation of the Toyota Production System (TPS).	01
	1.13	What is the alternate name for future state value stream map?	01
	1.14	Sketch the H-K planning model.	02
	1.15	What is Poka-Yoke?	02
		PART-B	
2	a	Explain the two pillars of Toyota Production system.	08
	b	Provide comparative analysis of mass production system and Toyota production system considering any four aspects.	08
		OR	
3	a	"The supply of value to the customer is a revolutionary concept as used in the TPS". Justify the statement.	08
	b	Outline the four basic concepts, lacking any of these, an enterprise is poorly suited to use Lean as a primary business philosophy.	08

4	a	Enumerate the six rules of kanban management and discus the unique functions of these rules.	08
	b	Critically examine the factors that cause the need for the three types of inventory in an organization.	08
		OR	
5	a	Explain in detail the method of calculating KANBANs.	08
	b	List and discuss the tactics, and skills required for achieving quantity control under the Jidoka pillar.	08
6	a	"Reducing the lead time is not only a result of reducing variation; it is a variation reduction technique of extreme power". Comment.	08
	b	What are the three fundamental issues must be addressed, any time any major culture-changing initiative is undertaken, such as the implementation of a lean initiative? Evaluate.	08
		OR	
7	a	As an enlightened manager highlight the cultural issues that you may have to address while implementing lean in an organization.	08
	b	An organization plans manufacturing spread over two ten-hour shifts and each shift includes a 30-minute launch and two ten-minute breaks. The normal work schedule is five days per week and there are nine holidays. The customer has a contractual agreement to purchase 500,000 units per year. Compute the takt time.	04
	c	Discuss the importance of one of the four key strategies part of the overall approach to make a value stream lean.	04
8	a	In what situations you recommend the following lean implementation approaches? Illustrate with reasons. i. Clean sweep ii. One-value-stream-at-a-time	08
	b	Evaluate the five procedures to implementing a lean initiative in an organization.	08
		OR	
9		Write notes on the following lean techniques: i. Single minute exchange of dies. ii. Present state value stream map. iii. Future state value steam map. iv. Spaghetti diagram.	16
10	a	Why are goals and goal development so important? Discuss.	08
	b	Explain the purposes and characteristics of goal that are developed as part of policy deployment.	08
		OR	
11	a	The ultimate purpose of transparency, starting with situation assessment is rapid response PDCA. Elaborate.	06
	b	Discuss the five step prescription on how to sustain the gains from lean implementation.	10

Total Quality Management

VI Semester SEE Question Papers

Course codes: 21IM65E2, 12IM6C2

- Sep/Oct 2024 (21IM65E2)
- Fast Track Jul 2019 (12IM6C2)
- Fast Track Jul 2018 (12IM6C2)
- Apr/May 2018 (12IM6C2)
- Fast Track Jul 2017 (12IM6C2)
- Apr/May 2017 (12IM6C2)
- Apr/May 2016 (12IM6C2)

21IM65E2

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Examinations Sept/Oct – 2024

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Expand the acronym PDCA.	02	1	1
	1.2	List the three elements of Joseph Juran's Quality Trilogy.	02	1	1
	1.3	Distinguish between Market-in and Product- out concepts.	02	2	2
	1.4	List the seven management tools that help companies identify latent needs and translate those needs into plans for products.	02	1	2
	1.5	Mention the two major ideas on which the concept of continuous improvement is based.	02	1	3
	1.6	WV model illustrates three types of problem solving. What are they?	02	1	3
	1.7	List the three phases in Hoshin Management.	02	1	1
	1.8	Expand the acronym DMAIC in six sigma context.	02	1	1
	1.9	What is meant by Lean Six sigma?	02	2	4
	1.10	Write a note on 'Rolled Throughput Yield' as a metric for six sigma.	02	2	4

PART-B

2	a	Illustrate and explain the evolution and development of Four Fitnesse's of Quality.	10	3	1
	b	Enumerate Philip Crosby's Four Absolutes.	06	2	1
3	a	"The market-in concept includes the idea of a process for improvement aimed at continuing customer satisfaction"- Analyze the statement.	10	4	2
	b	Illustrate and explain the elements of "management by process".	06	3	2
		OR			
4	a	"WV model illustrates the interplay between thought and experience"- Justify the statement.	10	4	3
	b	Distinguish between 'Reactive Improvement' and 'Proactive Improvement' methodologies in Total Quality Management.	06	2	3
5	a	"Joining of daily work and improvement work everywhere in the organization is the idea of dual function of work". Illustrate and explain.	06	3	2
	b	Summarize the seven principles for activating teamwork in organizations.	10	2	2

		OR			
6	a	Enumerate the seven elements of organizational infrastructure for implementing Total Quality Management.	10	2	2
	b	"Total Quality Management as a learning system aims at developing individual, team, company and national skill". Elaborate.	06	3	2
7	a	Examine different levels of expertise in terms of belts and their contribution to the implementation of six sigma.	10	4	1
	b	Write a note on the critical to Quality concept in six sigma.	06	2	2
		OR			
8	a	Enumerate any five benefits six sigma can offer to organizations.	06	2	2
	b	Performance metrics are critical for measuring and improving process performance in six sigma. In the above context discuss the relevance of the following: (i) Process capability indices. (ii) First pass yield. (iii) Cost of poor quality. (iv) Mean time between failures.	10	3	2
9	a	A leading automotive parts manufacturer is facing problem of high defect rates in the assembly line, leading to increased costs and customer dissatisfaction. Which six sigma methodology the organization should use to reduce the defects and improve the customer satisfaction? Discuss the methodology in detail in the context of above case.	16	5	4
		OR			
10	a	There are two types of problem solving methodology in six sigma. What are they? Also, discuss when to use each of those methodologies.	10	4	4
	b	Summarize the benefits of Lean six sigma to organizations.	06	2	4

12IM6C2

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Fast Track Examinations July-19

Electrical and Electronics Engineering

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from PartA. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	When should Six Sigma be used?	02
	1.2	Provide any four obstacles associated with TQM implementation.	02
	1.3	How are cause and effect diagrams useful?	02
	1.4	The main contribution of Masaki Imai is the concept of _____.	01
	1.5	The fundamental focus of the Quality of design as a concept is to define different _____ of quality for achieving the same functional requirement of the customers.	01
	1.6	As per the 6th principle for activating team work(Create an environment for team learning). Draw the sketch of physical arrangement for setting up meetings and justify.	02
	1.7	The major philosophical change between the 'market in' concept and the 'customer in' concept is to put the emphasis on _____ aspect of quality.	01
	1.8	Among the four fitness of quality; which fitnesses aim towards meeting quality of conformance and which ones for quality of design.	02
	1.9	A project that is $SMART$ is _____, _____, _____, _____, _____.	02
	1.10	Rewrite the following problem statement by applying guidelines for "Clear statement of theme". Understand customer deliver, quality and pricing requirement.	02
	1.12	The concept of _____ emphasizes the participation of all the employees in the organization is the quality improvement initiatives taken up by the top management.	01
	1.13	What are the three main kinds of teams as per TQM?	02
		PART-B	
2	a	Compare and contrast quality of conformance and quality of design. Illustrate with examples.	08
	b	Summarize Deming's approach to quality management in manufacturing industries.	08
		OR	
3	a	'Juraninsists that Quality does not happen by accident'. Identify the Quality planning roadmap as proposed by Juran and the other two aspects of Juran'strilogy namely Quality Control and Quality improvement. Give an example to clarify the three aspects of Juran trilogy.	08
	b	Explain in detail the quality concepts that characterizes the eras of fitness to standard and fitness to use.	08

4	a	What is semantics? How semantics helps in bridging the dual function of language? Elaborate.	08
	b	Using WV model, illustrate the three types of problem solving.	08
		OR	
5	a	Critically examine the steps in constructing the Affinity diagram.	08
	b	Shewart, Deming, Ishikawa and other practitioners of Statistical Process Control have proposed a model for Process Control. Describe this model for process control and mention the tools that help in execution of this model in the shop floor Quality context.	08
6	a	As a team lead of 10 members, how would you conduct meetings so that they are very effective?	04
	b	Analyze the importance of CEO involvement in TQM mobilization. Elaborate Domino theory of implementation.	08
	c	What is Hoshin Management? What are the phases in Hoshin Management?	04
		OR	
7	a	Bring out the comparison between Management by objectives and Hoshin Management. How does these two approaches distinguish the American and Japanese philosophers to Quality Management?	08
	b	Identify the elements of infrastructure for networking. Elaborate.	08
8	a	Take the example of Piston-Cylinder assemblies in Engines and identify the critical to Quality parameters. What are the consequences of process capabilities of the piston and cylinder manufacturing processes in this context.	08
	b	What are the common myths about six sigma? Why do you call them myths?	08
		OR	
9	a	List and explain the guidelines that helps to set up, measure and get results from the metrics.	08
	b	Explain what do you understand by the term "Process in Statistical Control". How does this reflect the stability of the manufacturing process in the mass production of components and sub-assemblies? Give a quantitative feel in the answer.	08
10	a	Evaluate $DMAIC, DFSS, DMADV$ methodologies with illustrations.	08
	b	Critically analyze the role of the Measure phase of the $DMAIC$ project methodology of the Six Sigma quality improvements. What are the measurement tools commonly used in $DMAIC$?	08
		OR	
11	a	Develop the framework for evaluating and prioritizing six sigma projects. Highlight the project selection criteria and significance ratings.	08
	b	On what basis a good lean six sigma project is identified?	08

12IM6C2

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VISemester B. E. ExaminationsFast Track July-18

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Fitness of cost means high _____ and _____ cost.	01
	1.2	Fitness to use and fitness to latent requirements have to do with _____ integration of the organization.	01
	1.3	Continuous improvement is based on _____ improvement and _____ improvement.	01
	1.4	Give a reason for presenting and diagnosing the quality improvement stories.	01
	1.5	Name the type of team that has a larger inherent risk of conflict than other types of teams.	01
	1.6	Hoshin management is also known as _____.	01
	1.7	Answer in one line: Why six sigma is known as a problem solving methodology?	01
	1.8	For a process of six sigma quality level the value of C_p =_____.	01
	1.9	A project selection matrix is a _____ and _____ decision matrix.	01
	1.10	What are the perspectives used to derive the basic criteria for selecting a six sigma project.	01
	1.11	Define quality of conformance.	02
	1.12	Illustrate with a sketch the optimum balance between too little and too much process.	02
	1.13	Meetings are important part of team learning. List substitutes for meetings.	02
	1.14	Provide a list of topics for training champion under six sigma.	02
	1.15	What is growth management system?	02

PART B

2	a	Explicate Dr. Armand V Feigenbaum's concept of total quality control.	06
	b	Trace the evolution of quality concepts. Elaborate the concepts and cite examples.	10
		OR	
3	a	Write a detailed note on Ishikawan's concept of company wide quality control.	06
	b	What are future fitnesses? Elucidate the importance in the Indian context.	06
	c	Compare and contrast quality conformance and quality of design.	04

4	a	Enumerate the key features of market in concept.	06
	b	The team leader of a quality improvement team is currently leading the team on a reactive improvement project. Suggest suitable tools and techniques for data collection, data analysis and cause analysis. You are also required to educate the team on the tools.	10
		OR	
5	a	List the general guidelines for managers diagnosing a quality improvement story. How do these guidelines help the QI team in developing problem solving skill?	06
	b	An entrepreneur in collecting data for a proactive improvement project. Evaluate and suggest the semantics methods for the purpose.	10
6	a	Evaluate the types of work groups in Japanese companies and three kinds of teams as per TQM.	06
	b	Develop strategy for TQM introduction for a small scale manufacturing company. The strategy may be based on the learning's from the CQM companies: Teradyne, BBN and digital.	10
		OR	
7	a	Write anHoshin indicating your goal for next year. Illustrate the alignment achieved by Hoshin management.	08
	b	Identify six elements of the infrastructure for networking. Develop a block diagram illustrating the interaction of these elements.	08
8	a	Analyze the costs of six sigma implementation.	04
	b	Elaborate the importance of assessing readiness of a company for implementing six sigma.	04
	c	Six sigma is about techniques and tools and statistics – but the results of six sigma depend on the people applying the tools etc. Write notes on the key roles played by executive leaders and the champions.	08
		OR	
9	a	Develop a plan for the six sigma initiatives in an organization. Detail the communication initiative, deployment plan, training for members and projects.	08
	b	Explain the guidelines to set up the proper metrics for six sigma implementation.	08
10	a	List the four aspects that make a project a poor choice with regards to its overall impact on business. Provide critical analysis of the above aspects.	08
	b	Develop a project selection matrix for evaluating and prioritizing projects.	08
		OR	
11	a	Provide a summary of DMA DV method of design for six sigma.	08
	b	Illustrate with a sketch the basic concept of growth management system.	08

12IM6C2

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Juran has made the greatest contribution to the management literature of any quality professional. He is altogether the author of _____ books and his _____ book has been translated into 13 languages.	01
	1.2	The fitness _____ requirement has made the concept of quality multidimensional.	01
	1.3	Who modified Kawakita's 'W' model for application to TQM and developed 'WV' model?	01
	1.4	A quality circle team is solving a quality problem. The team has to establish relationship between two parameters influencing the process. Suggest a suitable tool for the purpose.	01
	1.5	What is dual function of work?	01
	1.6	_____ are catalysts and one of the three elements necessary for successful societal diffusion of TQM.	01
	1.7	Six sigma is a program that focuses on the control of a process to the point of _____ from a center line.	01
	1.8	Define first time yield.	01
	1.9	The fundamental criterion for selection of six sigma projects is the opportunity to turn the _____ into the bottom line savings.	01
	1.10	_____ barrier is the point at which the efforts to improve products and services are limited by the capability of designs for products, services and processes.	01
	1.11	Among the four fitness of quality; which fitnesses aim towards meeting quality of conformance and which ones for quality of design.	02
	1.12	Rewrite the following statement with weakness orientation "Understand customer quality requirement".	02
	1.13	What are the three main kinds of teams as per TQM?	02
	1.14	Mention two important roles in which the master black belts are involved in an organization.	02
	1.15	List two reasons that companies give for the failure of new products.	02

PART B

2	a	Summarize Deming's approach to quality management in manufacturing industries.	08
	b	During the evolution of quality concepts, how did various organizations achieve fitness of cost? Elaborate with a detailed discussion.	08
		OR	

3	a	Develop the Juran's trilogy diagram. Explain the three phases with examples.	08
	b	Offer your comments on the following. i) Quality of conformance versus quality of design. ii) From devaitions to weakness to opportunities.	08
4	a	Illustrate the three types of problem solving with a sketch of 'WV' model. Write a note on process control.	08
	b	What is semantics? How semantics helps in bridging the dual function of language? Elaborate.	08
		OR	
5	a	Explain the market-in concept. How is it different from product-out concept?	08
	b	Select a problem of your choice. Illustrate the application of $7QC$ tools and $7QC$ steps.	08
6	a	Explain the seven principles for activating teamwork.	12
	b	What is Hoshin Management? What are the phases in Hoshin Management?	04
		OR	
7	a	Explain in detail, the Teradyne strategy for TQM introduction.	08
	b	What are the elements that are necessary for successful diffusion of TQM? Develop the sketch illustrating infrastructure for networking and explain.	08
8	a	"There are many misunderstandings about six sigma." List the myths and elaborate the truth.	08
	b	Explain the following in the context of using statistics in six sigma approach to management: i) Causes of variation. ii) Process capability.	08
		OR	
9	a	Develop a deployment plan for implementing six sigma approach in a medium scale manufacturing organization. List the elements of the plan and discuss.	08
	b	Elucidate the basic steps in setting metrics for an organization.	08
10	a	What are the perspectives that guide an organization in developing the list of possible projects for six sigma? Critically analyze.	08
	b	Provide an overview of $DMADV$ method of design for six sigma. Mention two differences between $DMADV$ and $DMAIC$.	08
		OR	
11	a	Develop the framework for evaluating and prioritizing six sigma projects. Highlight the project selection criteria and significance ratings.	08
	b	Identify the fundamental building blocks of growth management system. Elaborate the importance of knowledge management and lean six sigma.	08

12IM6C2

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Japanese companies worry about _____, not _____.	01
	1.2	_____ indicates the degrees to which a product or service confirms to standard.	01
	1.3	Poincare said "it is by logic we prove, but it is by _____ we discover.	01
	1.4	Two essential components to a TQM organization: _____ and _____.	01
	1.5	What is Hoshin management?	02
	1.6	Innovation is our competitor advantage, and process will stifle innovation. Comment.	02
	1.7	List the future fitness of quality.	02
	1.8	What are the four levels of TQM practice implicit in the four revolutions of management thinking?	02
	1.9	Differentiate between $SDCA$ and $PDCA$ cycle.	02
	1.10	What is product out concept?	02
	1.11	What is the purpose of using affinity diagram?	02
	1.12	List the myths about six sigma.	02

PART-B

2	a	Discuss how Juran's quality triology provide a successful frame work for achieving company's quality objectives.	10
	b	Bring out the contributions of Imai for Kaizen management practices.	06
		OR	
3	a	Illustrate the key features and weaknesses of the four fitnesses of quality.	06
	b	Each company must find its own way to implement TQM, but four areas of concepts and practices are common to most successful implementations, they represent the four revolutions in management thinking. Elaborate the four revolutions.	10

4	a	Explain how market-in concept helps the companies to focus on customer satisfaction as the purpose of work.	06
	b	WV model of continuous improvement guides in doing things right. Discuss the three types of improvements through the WV model.	10
		OR	
5	a	Discuss with case examples the seven steps in reactive improvement.	11
	b	Explain JiroKawakita's five principles for collecting data for proactive improvement.	05
6	a	Teams and team work are fundamental components of the way TQM organizes work. Comment on this statement.	09
	b	Explain briefly with example the 7 principles for activating team work.	07
		OR	
7	a	'Goals must be set for the company's TQM implementation and for the company's business'. Comment.	10
	b	Elaborate on the six elements of infrastructure for networking.	06
8	a	Discuss Six Sigma concept.	10
	b	Give an example to explain CTQ.	06
		OR	
9	a	Explain briefly Dos and Don'ts while implementing six sigma.	08
	b	State the benefits and draw backs of following six sigma methodology.	08
10	a	Explain $DMAIC$ approach to six sigma. List various tools used in each phase of $DMAIC$ cycle.	10
	b	How to sustain six sigma in an organization? Discuss.	06
		OR	
11	a	With a case example, explain the $DFSS$ methodology.	10
	b	Implementing lean six sigma approach has become a necessity in today's business environment. Justify.	06

12IM6C2

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	List any two deadly sins and diseases advocated by W Edwards Deming.	02
	1.2	"Japanese companies worry about cost, not process". Justify.	02
	1.3	The TQM slogan is, "Next process is your _____".	01
	1.4	Customer needs determine the _____ output whereas the process used determine the _____ output.	01
	1.5	Rewrite the following problem statement by applying guidelines for "Clear statement of theme". Understand customer deliver, quality and pricing requirement.	02
	1.6	As per the 6th principle for activating team work(Create an environment for team learning). Draw the sketch of physical arrangement for setting up meetings and justify.	02
	1.7	Write full form of the acronym $MITI$.	01
	1.8	At each lower level the Hoshins have the same format but are more _____.	01
	1.9	Define six sigma.	02
	1.10	Critical to quality concept in six sigma allows you to focus on improving quality from the perspective of the _____.	01
	1.11	List a benefit of six sigma that do not translate into Rupees, at least not directly.	01
	1.12	A project that is $SMART$ is _____, _____, _____, _____, _____.	02
	1.13	$DFSS$ initially requires a driving impulse from the _____ level in an company, well beyond verbal support.	01
	1.14	Six sigma enhances _____ and lean improves _____ of production processes in an organization.	01

PART B

2	a	Elucidate the salient features of Ishikawa's company-wide quality control.	08
	b	Summarize Philip Crosby's diagnosis repot of a troubled company. List the Crosby's absolutes for Quality Management.	08
		OR	

3	a	Explain in detail the quality concepts that characterizes the eras of fitness to standard and fitness to use.	08
	b	"Although each company must find its own way to implement TQM, four areas of concepts and practices are common to most successful implementations and these differ from practices in many non-TQM companies". Examine the statement, identify the four revolutions and discuss.	08
4	a	"TQM can be thought of as management in the face of rapid change". Illustrate with the help of a block diagram and highlight the importance of market-in concept.	08
	b	Compare the following: i) Process control and process improvement ii) Process versus creativity.	08
		OR	
5	a	Sketch the model of the steps in reactive problem solving. Illustrate the steps in identifying the problem with examples.	08
	b	Critically examine the steps in constructing the Affinity diagram.	08
6	a	Analyze the importance of CEO involvement in TQM mobilization. Elaborate Domino theory of implementation.	08
	b	Write brief notes on the three typical phases of TQM phase-in.	08
		OR	
7	a	"Hoshin management has three alignment purposes". Evaluate the statement and illustrate the three alignments.	08
	b	Construct the TQM model for skill development. Compile information about the complementary elements of indirect and direct learning.	08
8	a	Examine the six sigma methodology that helps to focus on engaging people and changing processes.	08
	b	Investigate why six sigma is not another quality program. Identify the myths and misunderstandings about six sigma.	08
		OR	
9	a	List and explain the guidelines that helps to set up, measure and get results from the metrics.	08
	b	Enumerate and discuss the basic steps in setting metrics.	08
10	a	Outline the $DMAIC$ methodology of six sigma approach to achieve process excellence.	08
	b	Evaluate $DMAIC, DFSS, DMADV$ methodologies with illustrations.	08
		OR	
11	a	Design a strategy for implementing six sigma in an organization.Summarize the steps involved.	08
	b	Present and defend your opinion on application of lean and six sigma methodologies for effectively managing the supply chains.	08

12IM6C2

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	Deming's contribution to the Quality Management is recognized through the use of _____ cycle for continuous Quality Improvement process.	01
	1.2	Juran's trilogy has three main aspects of _____ Quality Control and Quality Improvement.	01
	1.3	The main contribution of Masaxi Imai is the concept of _____.	01
	1.4	The fundamental focus of the Quality of design as a concept is to define different _____ of quality for achieving the same functional requirement of the customers.	01
	1.5	The term focus on customers as one of the four major revolution in management thinking refers to identifying the customer's _____ in a scientific process of capturing the customer's expectation from product or service offerings.	01
	1.6	The two types of improvement methodologies commonly adapted in Quality Management processes are Reactive Improvement and _____ Improvement.	01
	1.7	While the seven basic tools for Quality Control are called the seven QC tools, the seven tools used in the planning and Management of Quality are called the _____.	01
	1.8	Shoji Shiba –A Quality Guru has proposed a model for continuous improvement known as the _____.	01
	1.9	The identification of the Quality gaps in the product or process designs, provides an opportunity for Quality teams to initiate the process of _____, to improve the quality goals attained.	01
	1.10	The major philosophical change between the 'market in' concept and the 'customer in' concept is to put the emphasis on _____ aspect of quality.	01
	1.11	The concept of _____ emphasizes the participation of all the employees in the organization is the quality improvement initiatives taken up by the top management.	01
	1.12	Hoshinmanagement is a technique which emphasizes the _____.	01
	1.13	Define Societal Networking.	01

	1.14	One of the cultural aspects of Quality Improvement projects, when the Quality teams are engaged in problem solving in the contribution to _____ learning among the team members.	01
	1.15	The concept of $DPMO$ in the context of Six Sigma project stands for _____.	01
	1.16	The fundamental difference between the process capability approach to Quality of rejections and rolled throughput yield approach to Quality is in the _____ of the first pass yield, without the need for segregation according to the specification targets.	01
	1.17	$DFSS$ is an acronym which stands for _____.	01
	1.18	The phase in which the improvements, are instituted in the system as part of $DMAIC$ cycle and sustained is the _____ phase.	01
	1.19	The $DMADV$ method is the design approach to Six – Sigma is characterized by the D and v phase which stands for _____ and verify part of Quality Improvement projects.	01
	1.20	The lean six sigma approaches are driven by the simultaneous attack on the process inefficiencies and focus on _____ improvements in processes generating the product features.	01

PART-B

2	a	Trace the evolution of the Quality concept using the four fitnesses. Briefly characterize the concept of Quality as enshrined in the four quality concepts.	08
	b	Distinguish between Quality of Design and Quality of Conformance. Take the example of a Mobile handset and the features of a smartphone to bring out the distinction between these two aspects of Quality.	08
		OR	
3	a	Examine the Demings 14 points for Management. List the 14 points with a very brief explanation of each of these points with a link to Quality initiatives and the responsibility of the Top Management.	08
	b	'Juran insists that Quality does not happen by accident'. Identify the Quality planning roadmap as proposed by Juran and the other two aspects of Juran's trilogy namely Quality Control and Quality improvement. Give an example to clarify the three aspects of Juran trilogy.	08
4	a	Distinguish between the product – out concept and the Market – in concept. Why is the market – in necessary in the face of rapid change? Give an example.	08
	b	Justify the use of the $W-V$ Model for problem solving. How does this model help organizations to be guided through the three types of improvement?	08
		OR	

5	a	Shewart, Deming, Ishikawa and other practitioners of Statistical Process Control have proposed a model for Process Control. Describe this model for process control and mention the tools that help in execution of this model in the shop floor Quality context.	08
	b	Design the scenarios which promotes the interaction between the $PDCA$ cycle and the $SDCA$ cycle. Graphically depict the interaction between the $SDCA$ cycle and the Reactive Improvement Cycle.	08
6	a	Teams and Team work are a fundamental component of the way TQM organizes work. Explain as to why these components are important for Quality Improvement projects in the face of complexity in business.	08
	b	Cross functional teams involved in Quality Improvement projects using the TQM approaches can benefit by treating TQM as a learning system and TQM Model for skill development. Take an example of improvements on an existing product or service solution and explain the benefits for the teams.	08
		OR	
7	a	What is Hoshin Management? List the three alignment purposes Hoshin Management. Illustratively depict the alignments in Hoshin Principles.	08
	b	Bring out the comparison between Management by objectives and Hoshin Management. How does these two approaches distinguish the American and Japanese philosophers to Quality Management?	08
8	a	How can any organization assess the readiness for the Six Sigma programs? What are the steps in planning for Six Sigma?	08
	b	Take the example of Piston-Cylinder assemblies in Engines and identify the critical to Quality parameters. What are the consequences of process capabilities of the piston and cylinder manufacturing processes in this context.	08
		OR	
9	a	Explain what do you understand by the term “Process in Statistical Control”. How does this reflect the stability of the manufacturing process in the mass production of components and sub-assemblies? Give a quantitative feel in the answer.	08
	b	Distinguish between baselines and Benchmarking as Quality Concepts. How do these concepts drive Quality Improvement Projects?	08
10	a	Critically analyze the role of the Measure phase of the $DMAIC$ project methodology of the Six Sigma quality improvements. What are the measurement tools commonly used in $DMAIC$?	08
	b	How can one assess the business impact of a Six Sigma project? Give an example to illustrate the need to attach priority to selecting projects with high business impacts.	08
		OR	

11	a	"The Goal of Lean is elimination of waste and the goal of Six Sigma is to minimize defects". How can these two goals be combined using Lean Six Sigma project framework? Explain with an example.	08
	b	How can large organizations help its suppliers in Quality Improvement Processes? Why is it necessary to focus on supplier's Quality program in the Supply Chain context?	08

Design of Experiments

VI Semester SEE Question Papers

Course codes: 18IM6D4, 16IM6D3

- Sep 2023 (18IM6D4)
- Apr/May 2019 (16IM6D3)

18IM6D4

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations September-2023

Industrial Engineering and Management

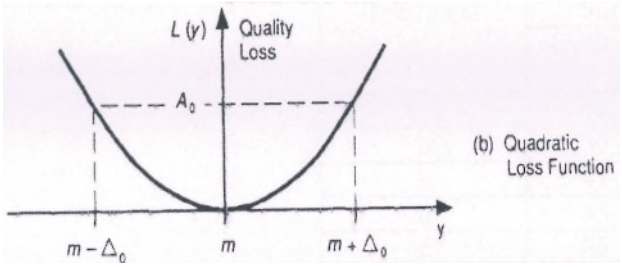
DESIGN OF EXPERIMENTS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	State the major disadvantage of the $OFAT$ strategy.	01
	1.2	_____ developed the foundations of robust design and validated its basic philosophies by applying them in the development of many products.	01
	1.3	_____ table is used to modify the Standard linear graph.	01
	1.4	What is the meaning of the notation:$L162^{15}$.	01
	1.5	What is principal fraction with reference to fractional factorial designs?	01
	1.6	Which is the design that has highest resolution among the Resolution III, IV and V designs?	01
	1.7	Select a manufacturing process of your choice; identify controllable factors, noise factors and response.	02
	1.8	List the three steps strategy that has as intuitive appeal for the engineering problem of optimizing a product or process design.	02
	1.9	Sketch the general model of a process.	02
	1.10	Identify the quadratic loss function plotted in the Fig 1.10. Write the expression for the same.  Fig 1.10	02
	1.11	Illustrate with a sketch the treatment combinations in the 2^2 design.	02
	1.12	Suppose the functional limits for the colour density of television sets are m$\pm$7. Let the repair of a television set in the field cost \$98. What is the average quality loss incurred by the customers receiving sets with colour density m+4.	02
	1.13	Develop the required linear graph for a situation in which there are four factors $(A, B, C \,\&\, D)$ at two levels each. All the main effects and interaction effect AB and CD are to be estimated.	02

PART B

2	a	Explain with examples the three basic principles of experimental design.	08
	b	Classify the noise factors and explain with a product of your choice.	08
3	a	How do you carry out data analysis from a single replicate of 2^k design? Discuss.	04
	b	A chemical engineer has conducted an experiment to study the effect of temperature and pressure on the reaction time of a chemical process. Let the two factors temperature and pressure are denoted by A and B respectively. Suppose the following results are obtained from this study. The number of replications is two.	

$Pressure\ (B)$	$Temperature\ (A)$	
	Low (60°C)	$High$ (70°C)
Low (100 MPa)	40	43
	37	50
$High$ (150 MPa)	59	37
	54	43

i) Estimate the factor effects. Which effects are large?

ii) Perform an analysis of variance, what are your conclusions? 12

OR

4 An experiment was conducted to study the effect of tool type (A), speed (B) and feed (C) on the surface finish of a machined part. Each factor was studied at two levels and obtained two replications. The order of experiment was random. The results obtained ate given in Table.

$Treatment$	$Surface\ Finish$	
	$Replicate\ 1$	$Replicate\ 2$
(1)	54	73
a	60	53
b	41	51
ab	43	49
c	86	66
ac	82	73
bc	63	65
abc	66	65

i) Estimate the factor effects. Which effects are large?

ii) Carryout analysis of variance, what are your conclusions? 16

5	a	Illustrate with an example confounding 2^3 factorial design in to 2 blocks.	06
	b	Explain with an example partial confounding	05
	c	Write a note on confounding 2^k factorial design in four blocks.	05
		OR	
6	a	The successful use of fractional factorial designs is based on three key ideas. Elaborate.	06
	b	Discuss the importance resolution III, IV and V experimental designs.	06
	c	Write a note on one quarter fraction of the 2^k design.	04

7	a	Explain with illustration the Dummy level technique.	06
	b	Consider an experiment with four factors $(A, B, C \text{ and } D)$ each at two levels. Also, the interactions AB, AD and BD are of interest to the experimenter. Design the Orthogonal Array experiment layout.	10
8	a	List the steps in robust design.	06
	b	Explain any four signal-to-noise ratios for static problems. Illustrate with sketches.	10

16IM6D3

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VISemester B. E. Examinations April/May-19

Industrial Engineering and Management

DESIGN OF EXPERIMENTS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical tables and orthogonal array tables permitted.

PART A

1	1.1	In the early years of statistically designed experimentation the ideas of efficient design for conducting research was applied to the field of agriculture. What was the performance measure in this case?	01
	1.2	Sketch the Ideal Curve for the quality loss function.	01
	1.3	The total number of treatment combination when 4 factors are investigated each at 2 levels will be?	01
	1.4	Where in the P diagram do you indicate the Noise factors?	01
	1.5	Mention any one merits of the factorial experimentation.	
	1.6	What is the significance of listing treatments as per Yate's order? (standard order).	01
	1.7	If there is a factor A at 3 levels and factor B at 3 levels, what will be the degree of freedom for the interaction term AB?	01
	1.8	Mention any one benefit of representing the results of a technical experimentation using $ANOVA$ table.	01
	1.9	What is the advantage of the blocking as a strategy of experimentation?	01
	1.10	If effects are confounded, then what would be the impact on the analysis of main effects and interaction effects?	01
	1.11	Define a Resolution III design precisely.	01
	1.12	What is the advantage of a fractional factorial design over a full factorial design for experimentation in R & D situations?	01
	1.13	What is the meaning of L_9^{34} notation?	01
	1.14	Give the typical structure of a Linear graph?	01
	1.15	Do you agree that the interaction table and Linear graph serve the same purpose while constructing orthogonal array for a experimental situation.	01
	1.16	Which technique is most suited for accommodating a factor with 4 levels in a 2 level orthogonal array layout design?	01
	1.17	Write the S/N ratio for Nominal the best quality characteristic type of problem.	01
	1.18	What is an outer array?	01
	1.19	Graphically represent the concept of robustness through the principle of exploiting the non-linearity of the responsive curve (input parameters versus response characteristic curve).	01
	1.20	What does Grey denote in the Grey Taguchi technique?	01

PART B

2 a "Offline methods of quality improvement and online methods of quality control need to be initiated to take care of the product and process quality requirements in organization". Elaborate on the principles, approaches and techniques related to online and offline quality control and improvement with examples. 08

b A TV manufacturer has observed that the average cost to repair the deviation in a particular parameter X is $Rs.1500$. If the parameter X has a tolerance Zone as $50 <= x <= 90$, with the target as 70. Derive the corresponding Quadratic loss function.
For the above situation, estimate the average loss for $5\,TV$ sets with measurements for parameter X yielding 58,62,70,52,75. Sketch your results graphically. 08

3 a One of the critical measures of some raw paper is the ability to minimize the absorption of water for a specific time and then reweighing the sample. A percentage weight gain is the measure of the water absorption characteristic. In an effort to minimize the absorption, two factors have been selected for experimentation

Factor A- Type of sizing chemical used
Levels being normal and enhanced
Factor B- Amount of the sizing chemical Levels being 5% and 8%

A 2^2 factorial design was run. Four replicates were taken. The result is in the Table 3(a) below:

Table 3(a)- Data for example

		$\boldsymbol{B}(Sizing\ type)$			
$(B)Amount\ of$		$Normal(-1)$		$Enhanced(+)$	
	5%	1.34	5.24	10.38	6.66
	$(-)$	6.56	4.20	6.68	9.56
	8%	4.98	3.49	6.22	11.0
	$(+)$	3.92	5.29	4.72	8.36

i) Calculate the effects.
ii) Create pareto chart of the effects.
iii) Carry $ANOVA$ and identify the significant effects. 10

b Using the data from part (a) above, derive an appropriate regression model and compute the residuals. Comment on the residual values. 06

OR

4 a A mechanical engineer is studying the thrust force developed by a drill press. He suspects that the drilling speed and feed rate of the material are the most important factors. He selects four feed rates and uses a high and low drill speed chosen to represent the extreme operating conditions. He obtains the following results. Analyze the data and draw conclusions. Use $\alpha = 0.05$

	$Feed\ Rate$			
$Drill\ Speed$	0.015	0.030	0.045	0.060
125	2.70	2.45	2.60	2.75
	2.78	2.49	2.72	2.86
200	2.83	2.85	2.86	2.94
	2.86	2.80	2.87	2.88

10

	b	Write down the structure for the Analysis of Variance of a general 2^K design. What will be the total degrees of freedom?	06
5	a	What is the meaning of Confounding? Can there be partial Confounding as well? Justify.	06
	b	Construct a 2^{5-1} design. Show the design may be run in two blocks of eight observations each. Are any main effects or two factor interactions confounded with blocks?	10
		OR	
6	a	A 2^{5-2} design was run to investigate the effect of A-Condensation temperature B- Amount of material-1 C-Solvent Volume D-Condensation time E- Amount of material 2 The yield results obtained are as follows $e = 23.2 \;\; ad = 16.9 \;\; cd = 23.8 \;\; bde = 16.8$ $ab = 15.5 \; bc = 16.2 \; ace = 23.4 \;\; abcde = 18.1$ i) Verify the design generates used where $I = ACE$ and $I = BDE$ ii) Write down the complete defining relation and alias for this design iii) Estimate the main effects	10
	b	Explain the process of arriving at the resolution of a design. What is the meaning of a Resolution III, IV, V designs?	06
7	a	Outline the steps for the construction of Orthogonal arrays for experimental investigation with the help of an example of your own. How do you accommodate a three level factor in a 2 level design?	06
	b	In an Investigative study on the stability of power factor of capacitor, it was decided to examine the following factors: $Sl.No.$ | Factors | $Levels$ 1 | $Leakage\ Current\ of\ foil(A)$ | $<= 2mA \quad >= 2mA$ 2 | $Power\ Factor(B)$ | $<= 3\% \quad >= 3\%$ 3 | $Capacitance\ value\ of\ coil(C)$ | $8-10\mu f \quad 11-13\mu f$ 4 | $Soaking\ time\ during\ \ Impregnation(D)$ | $2hrs \quad 4hrs$ 5 | $Loading\ Pattern\ (E)$ | $70\% \quad 100\%$ Suspected interactions effects are AB, BC. Develop the Orthogonal Layout.	10
8	a	Write down the Signal to Noise ratios for any two cases of Static Quality characteristics and for any two cases of Dynamic Quality characteristics problems. Give examples for each of the above type of problems.	08
	b	Compute the Signal to Noise ratio corresponding to a Quality characteristic with 5 replication values as $1.0, 0.80, 0.92, 1.02$ and 0.83. Compute the S/N ratio under the following assumptions for the state problem with Quality characteristics being treated as: i) Smaller the better type ii) Nominal the best type iii) Larger the better type.	08

Multicriteria Decision Modelling

VI Semester SEE Question Papers

Course codes: 18IM6C3

- Sep 2023 (18IM6C3)
- Aug 2022 (18IM6C3)

18IM6C3

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations September-2023

Industrial Engineering and Management

MULTI CRITERIA DECISION MODELLING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	State the description for the terminologies- alternative, decision maker in decision making environment.	02
	1.2	What two requirements should a problem satisfy in order to be suitable for solving it by a Genetic Algorithm?	02
	1.3	What are the four types of $MCDM$ methods in Discrete Multicriterion Decision Making?	02
	1.4	What is the meaning of a deviational variable in Goal programming?	02
	1.5	What is the significance of 1,5,7,9 in Saaty's Scale?	02
	1.6	Mention any one of the techniques under the category of: a) Non-traditional optimization methods b) Traditional optimization methods.	02
	1.7	Expand the acronym $MAUT$.	02
	1.8	State the principle of $TOPSIS$.	02
	1.9	What range of values vary in degree of similarity and which is considered the best?	02
	1.10	What is Signal to Noise ratio? And state three possible types of computations in it.	02

PART B

2	a	What is structured decision making? Illustrate the structured decision making situations in the field of engineering and management with an example.	06
	b	Outline the importance of simulated annealing and simulated quenching with a help of a flow chart.	10
3	a	Explain Multi-Objective Differential Evolution ($MODE$) with necessary steps.	08
	b	State some problem areas in management where goal programming might be applicable.	08
		OR	

4	a	List the methods falling in Plan Generation and Selection. Discuss any one.	08
	b	An office equipment manufacturer produces two types of products: chairs and lamps. The production of either a chair or a lamp requires one hour of production capacity in the plant. The plant has a maximum production capacity of 50 hours per week. Because of the limited sales capacity, the maximum numbers of chairs and lamps that can be sold are 6 and 8 per week respectively. The gross margin from the sale of a chair is $Rs.90$ and from the sale of a lamp is $Rs.60$. The plant manager desires to determine the number of units of each product that should be produced per week in consideration of the following set of goals. Goal 1: Available production capacity should be utilized as much as possible but should not exceed 50 hours per week. Goal 2: Sales of two products should be as much as possible. Goal 3: Overtime should not exceed 20% of available production time. Formulate this problem as a GP model so that plant manager may achieve his goals as closely as possible.	08
5	a	List the steps involved in Multicriteria Decision making ($MCDM$) methodology.	06
	b	Discuss the Analytical Hierarchy Process for a packaging problem as an example with a neat block diagram.	10
		OR	
6	a	Explain Composite Programming of distance based Multicriterion Decision Making Methods.	06
	b	Compute priority of alternatives for the payoff matrix presented in the following table using weighted average method. Assume weights for each criterion as $0.1, 0.12, 0.15, 0.20, 0.33$ and 0.1. Use normalization method 4 for analysis. $Alternate$ | $Criteria$: C_1 | C_2 | C_3 | C_4 | C_5 | C_6 A_1 | 8 | 9 | 6 | 5 | 2 | 4 A_2 | 6.5 | 5.8 | 2 | 9 | 6 | 3 A_3 | 8.3 | 6 | 4 | 7 | 4 | 4 A_4 | 4 | 8 | 8 | 7 | 3 | 8 A_5 | 5.2 | 7.2 | 4 | 2 | 1 | 6 A_6 | 9.4 | 9 | 6 | 3.2 | 2 | 4 Hint : Normalized method 4 $= \frac{f_j(a)}{\sqrt{[\sum_{a=1}^{N} f_j(a)^2]}}$	10
7	a	In your opinion which fuzzy number based approach is advantageous in decision making scenario? Triangular or trapezoidal? Why?	06

	b	Three infrastructure projects A_1 to A_3 are evaluated on four criteria namely C_1, C_2 C_3 and C_4. Evaluations are presented in table below as trapezoidal fuzzy numbers in the normalized payoff matrix. All the criteria are maximization in nature. Assume crisp weights of criteria as $0.10, 0.20, 0.25$ and 0.45. Determine which infrastructure project is the best. Explore three defuzzification methods.	10

	Criteria															
	$C_1(0.10)$				$C_2(0.20)$				$C_3(0.25)$				$C_4(0.45)$			
Alt	a_1	b_1	c_1	d_1	a_2	b_2	c_2	d_2	a_3	b_3	c_3	d_3	a_4	b_4	c_4	d_4
A_1	0.40	0.50	0.60	0.70	0.20	0.40	0.60	0.80	0.70	0.85	0.90	1.00	0.10	0.30	0.40	0.45
A_2	0.15	0.25	0.35	0.45	0.20	0.30	0.40	0.50	0.25	0.35	0.40	0.45	0.30	0.50	0.70	0.90
A_3	0.10	0.20	0.30	0.40	0.52	0.30	0.50	0.70	0.40	0.50	0.70	0.75	0.35	0.50	0.55	0.60

8	a	Describe Taguchi methodology and its role in Multicriteria Decision making.	06
	b	Five irrigation farms were analyzed for the cost per unit of groundwater pumping. Three variations of costs of groundwater pumping were found in each farm and presented in Table below.	

Irrigation farm	*Cost of groundwater for scenario*		
	1	2	3
1	270	290	285
2	300	320	240
3	280	260	270
4	210	230	320
5	280	295	240

Select which farm you prefer as pilot project for improving other farms. 10

18IM6C3

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations August-2022

Industrial Engineering and Management

MULTICRITERIA DECISION MODELLING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Define the term pay off matrix	01
	1.2	Mention any one difference between Multi criteria decision making and multi attribute decision making.	01
	1.3	Mention any of the techniques under the category of Non Traditional Optimization methods.	01
	1.4	The tem used for the initial population of a string of variables goes by which name (As applied to Genetic algorithm).	01
	1.5	Write down the mathematical expression for the weighted sums of various objectives for the weightage method.	01
	1.6	The Acronym MOGA has its full form as what?	01
	1.7	What is the meaning of a deviational variable in the context of Goal programming?	01
	1.8	Mention any two groups under the methods falling in the plan selection category as part of multi objective optimization.	01
	1.9	Define compromise programming.	01
	1.10	What kind of approach is adapted in the case of co-operative Game theory?	01
	1.11	Expand the acronym TOPSIS.	01
	1.12	Write down the expression for the separation measure Da^+ in f dimensional Euclidean distance for each alternative a	01
	1.13	Sketch a typical fuzzy number $\bar{A}$ with triangular membership function.	01
	1.14	Write the expression for relative closeness C_i of each alternate i with reference to negative ideal measure $D_{\bar{\imath}}$	01
	1.15	Fuzzy compromise programming is an extension of which technique in fuzzy environment.	01
	1.16	Expand the acronym PROMOTHEE.	01
	1.17	Can the Data envelopment analysis be considered as a special form of Linear programming? How?	01
	1.18	What is the central idea in the Taguchi methodology?	01
	1.19	Define Swarm Intelligence.	01
	1.20	Who is the propounder of particle swarm optimization?	01

PART B

2 a Explain the necessity of structured decision making? Outline the steps and methodical aspects of structured decision making. 08

b Solve the single objective LPP problem

$\text{Max}\, Z = 2x_1 + 3.5x_2$

$s/t \quad 4x_1 + 3x_2 \geq 6$

$5x_1 + x_2 \leq 1.5$

$x_1 \geq 1.5$

$x_2 \geq 1.0$ 08

3 a Two crops x_1 and x_2 are planned in a command area where water and fertilizer are major constraints whereas sufficient land is available for planning (ie land is not a constraint). Three objectives are the proposed for the command area planning scenario viz Net benefits, agricultural production and labour employment which are all of maximization in nature. Details of the crops such as net benefits (Rs/ha) , agricultural production (T/ha)., Labour employment (man-days/ha) , Fertilizer requirement (kg/ha), water requirement (ha-m/ha) and minimum bands on area under crops are presented in table below.

Various parameter of the planning problem

Characteristics	Crop x_1	Crop x_2	Resources
Net benefits	400	500	---
Agricultural production (T/ha).	2	5	--
Labour employment (man-days/ha)	10	5	--
Fertilizer requirement (kg/ha)	0.41	0.32	28kg
Water Requirement (ha-m/ha)	0.22	0.48	30 ha-m
Minimum bands on area under crops (ha)	25	30	

Solve the problem in multi objective framework using weightage method with weight sets (0.2,0.3,0.5) (0.4,0.2,0.4)(0.1,0.2,0.1) assigned to net benefits (B) , agricultural production (P) and Labour employment (L). Use normalization method for the analysis. 12

b Briefly highlight the ideas behind the multi objective genetic algorithms (Any 3 ideas central to its working would suffice). 04

OR

4 a Alpha company produces two kinds of fancy products, pen holder and paper trays. Production of either of them requires 1 hr. production capacity in the plant. The plant has a maximum production capacity of 12 hours per week. The maximum number of pen holders and paper trays that can be sold are 7 and 10 respectively. The gross margin from the sale of pen holder is Rs.90 and Rs.45 for paper tray. The plant manager has set the following goals in order of importance.

		P1: He wants to avoid underutilization of production capacity P2: He wants to limit overtime hours to 3 hrs P3: He wants to sell as many as pen holders and pen twice as possible. Since the gross margin from the sale of a pen holder is set as twice the amount of the profit from paper tray the manager has twice much desire to achieve the sales goals for pen holders as for paper trays P4: The manager wishes to minimize the overtime operation of the plant as much as possible Formulate the goal programming problem based on the above information	10
	b	Differentiate between (i) Goal programming ad Compromise programming (ii) Goal programming and fuzzy goal programming.	06
5	a	Tom schmid is thinking about buying a Nordic Sci machine. The three important features to him are price, Ease of use and the ability to store the exercise equipment in a closet when he is done using it. Given the following data, help Tom determine the best machine for him. <u>Factor Weights</u> Factor — Importance Weight Price — 0.9 Ease of Use — 0.75 Storage — 0.6 <u>Factor Evaluation</u> Factor — Professional Nordic Skier — Econo Nordic skier Price — 0.5 — 0.8 Ease of Use — 0.95 — 0.6 Storage — 0.9 — 0.7	08
	b	George is about to buy a compact stereo casette player. He is currently considering three brands Sun, Histen and Surgo. The three important factors to George are the price, color , warranty, size of the unit and the brand name. George has determined factor of weights of 0.4,0.1,0.1 0.1 and 0.3 respectively. Furthermore George has determined factor evaluations for all of the three different manufacturers of that unit he is considering. The sun unit has factor evaluation of 0.7,0.9,0.8,0.8 and 0.9 for the price , color warranty, size and brand name factors. The Hitek unit has factor evaluation of 0.6,0.9,0.9,0.8 and 0.9 of these factors. Finally Surgo has factor evaluation of 0.8,0.4,0.4,0.2 and 0.6. For the same features of price, color, warranty , size and brand name. Determine the total weighted evaluation for the three manufacturers. Which one should George select?	08
		OR	

6 a Compute the ranking pattern of alternatives given the following pair of matrix using the TOPSIS method

Criteria / Alternate	C_1	C_2	C_3	C_4	C_5	C_6
A_1	8	9	6	5	2	4
A_2	6.5	5.8	2	9	6	3
A_3	8.3	6	4	7	4	4
A_4	4	8	8	7	3	8
A_5	5.2	7.2	4	2	1	6
A_6	9.4	9	6	3.2	2	4

Assume the weights of the criteria as 0.1,0.12,0.15.0.20.0.33,0.10. Use normalization method 4 for analysis. 10

b Compute the ranking portion for the pay off matrix presented in the table below using compromise programming method for p=1,2. Assume weights for criteria as 0.35, 0.40, 0.25. All the criterion are of maximization in nature.

Criteria / Alternate	C_1	C_2	C_3
A_1	35.250	0.77718	1152.00
A_2	36.300	0.77250	1301.80
A_3	38.492	0.76386	1439.50
A_4	40.331	0.74119	1586.50
A_5	43.3239	0.66374	1599.40
A_6	45.500	0.58989	1478.70

06

7 Four infrastructure projects A1 to A4 are evaluated in fuzzy environment on three criteria namely net monetary benefits (C_1), social benefits (C_2) and environmental construction (C_3). All criteria are of maximization in nature. Evaluations are rated as triangular fuzzy number in the payoff matrix based on Chen's methodology. Assume weights of criteria as (0.25,0.30,0.35), (0.40,0.45,0.50) (0.50,0.55,0.60). Identify substitute infrastructure projects using fuzzy TOPSIS.

Criteria / Alternate	C_1			C_2			C_3		
	a_1	b_1	c_1	a_2	b_2	c_2	a_3	b_3	c_3
A_1	3	4	5	3	4	6	3	5	7
A_2	1	3	5	4	5	7	4	5	6
A_3	2	4	6	2	4	6	3	4	6
A_4	5	7	8	4	8	10	5	7	9

16

8 a Describe advantages and limitations of Data Envelopment Analysis (DEA). Compare Data Envelopment Analysis with multicriteria Decision making (MCDM). 08

b Differentiate Ant colony optimization , particle swarm optimization and Genetic algorithms on the bases of at least three characteristics. 08

Advanced Manufacturing Process

VI Semester SEE Question Papers

Course codes: 18IM6C5

- Aug 2022 (18IM6C5)

18IM6C5

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations August-2022

Industrial Engineering and Management

ADVANCED MANUFACTURING PROCESS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	In shell moulding sand preparation, the resin is combined with a catalyst (hexa) to develop the _____ characteristics.	01
	1.2	Dental castings and turbine blades are produced by _____ casting.	01
	1.3	In plaster mould castings, talc is added to the plaster while mixing to prevent _____ of mould.	01
	1.4	The thickness of shells in shell moulding is ranging from _____ to _____.	01
	1.5	Petrol engine cylinders are made by _____ moulding.	01
	1.6	Use of runner, risers and cores are eliminated in _____ casting.	01
	1.7	Strip drawing test is a method which is mainly used for evaluating the characteristics of _____.	01
	1.8	The flow stress _____ with strain rates during High Energy Rate Forming (HERF) Processes.	01
	1.9	Explosive forming is a metalworking technique in which an _____ charge is used instead of a punch.	01
	1.10	Metal forming is a process in which the metal is deformed _____ to get into the desired shape.	01
	1.11	A proper care should be taken while cooling or solidification of castings to avoid _____ formation.	01
	1.12	Rapid heat extraction from the casting causes narrowing of liquid-solid zone which sweeps itself across the _____ metal.	01
	1.13	An alloy, unlike a pure metal, does not have a sharply defined _____ temperature.	01
	1.14	Composite materials are classified based on the type of _____.	01
	1.15	Fibers or particles embedded in matrix of another material are the examples of _____ material.	01
	1.16	Fiber Reinforces Composites are composed of fibers embedded in _____ material.	01
	1.17	Cellular manufacturing is an approach whereby production can be done in _____ batches.	01
	1.18	Cellular manufacturing is also known as _____ Technology.	01
	1.19	The main objective(s) of Industrial robot is to _____ the labour requirement.	01

	1.20	The Robot designed with cylindrical coordinate systems has _____ linear and one rotational movement.	01

PART B

2	a	Differentiate between true centrifugal casting and semi centrifugal casting?	06
	b	Explain the process along with the principle involved in vacuum mould casting.	10
3	a	Discuss the process variables and advantages associated with explosive forming.	08
	b	Enumerate the importance of sticking friction in metal working and why is friction generally undesirable in metal forming operations?	08
		OR	
4	a	Discuss the different operations involved in bulk deformation process.	06
	b	Explain the principle and Process Characteristics of Electro hydraulic Forming.	10
5	a	Analyze the use of Computer Aided Engineering (CAE) concepts in a product manufacturing.	08
	b	List the application of CAE techniques in Thermal analysis of Heat-Affected Zone.	08
		OR	
6	a	Elaborate on the application of CAE technique in the analysis of forging process.	08
	b	Explain the role CAE techniques in the simulation of molten metal flow.	08
7	a	Explain the importance of injection moulding with respect to the following: i. Three-plate mould ii. Two-plate mould	08
	b	Explain the four major steps typically involved in the making of composite products.	08
8	a	What is a flexible manufacturing system and state the advantages of flexible manufacturing system technology, compared to conventional batch operations.	08
	b	Define group technology and explain the “composite part concept” in group technology.	08

Digital Manufacturing

VI Semester SEE Question Papers

Course codes: 18IM6D5, 16IM6C1

- Aug 2022 (18IM6D5)
- Apr/May 2019 (16IM6C1)

18IM6D5

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)
VI Semester B. E. Examinations August-2022
Industrial Engineering and Management

DIGITAL MANUFACTURING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	____ is a modeling tool applied in discrete asynchronous concurrent systems and ____ is a set of objects having same attributes and services.	02
	1.2	Reverse engineering is also known as _____.	01
	1.3	___ has become the indispensable drive factor in the product life cycle of the manufacturing industry.	01
	1.4	Basic modeling method is also called _____.	01
	1.5	What is initial stage of manufacturing where manufacturing existed as a skill?	02
	1.6	What is the relevance of LAN in Digital Manufacturing?	01
	1.7	Mention any two key technologies of DM.	02
	1.8	The middle layer in the hierarchical structure of intelligent sensing system is also called ____.	01
	1.9	Human factors engineering is termed as ____.	01
	1.10	Name any one hybrid diagram-Analysis model.	01
	1.11	_____ System is in a state of nonlinear system or nonlinear region.	01
	1.12	Simulating the manufacturing activity of human experts through computers is called _____.	01
	1.13	Cleaning the data to eliminate noise and inconsistencies is called ____.	01
	1.14	What is Metadata?	02
	1.15	Mention any two sensors deployed in machining process monitoring.	02

PART B

2	a	Elaborate on the key features of digital manufacturing system.	06
	b	Illustrate and Explain with a flow chart the Architecture in digital manufacturing system.	10
3	a	"Materialization of manufacturing information is a special transmission and transformation of original manufacturing information". Elaborate.	10
	b	Brief about the operation process of virtual manufacturing alliance organization with Process flow diagram.	06

		OR	
4	a	What is Virtual prototyping? Illustrate its frameworks.	10
	b	Explain the various applications of reverse engineering in digital manufacturing.	06
5	a	Explain the framework of functions to realize the autonomy of manufacturing system.	10
	b	Explain the process of data mining.	06
		OR	
6	a	Discuss the concepts and features of Intelligent manufacturing.	06
	b	Explain the general process to design the model of management of technology(MOT)	10
7	a	State the differences between CAPP/CAM.	06
	b	Explain how bottleneck problem of information resource sharing is done in digital manufacturing.	10
8	a	Explain the role of Digital manufacturing technology in Micro-Nano manufacturing.	06
	b	With the help of a sketch explain the relationship between the manufacturing digitalization and manufacturing green revolution.	10

16IM6C1

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

DIGITAL MANUFACTURING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	What are the "3*S*" of an enterprise to be analyzed before digitizing?	02
	1.2	What are the three changes of Digital Marketing?	02
	1.3	What are the components that perform as a package in Digital Marketing?	02
	1.4	What is metadata?	02
	1.5	List the different types of network environments needed in implementation of Digital Marketing.	02
	1.6	Wire frame modeling, surface modeling and Solid Modeling are _____ modeling forms.	01
	1.7	*VLSI* stands for _____.	01
	1.8	Manufacturing systems can be classified as _____ and _____ manufacturing systems.	02
	1.9	*WAN* stands for _____.	01
	1.10	Expand *MEMS*.	02
	1.11	List the technologies involved in Digital Recognition and Integration.	02
	1.12	What is extreme manufacturing?	01

PART B

2	a	What are the various stages of development of traditional, nontraditional and digital manufacturing systems?	08
	b	Illustrate the Iterate-Adjust-Optimize loop. Support your answer with suitable live example.	08
3		Highlight the foundation of digital manufacturing and list the modeling methods of digital manufacturing system and explain any one in detail.	16
		OR	
4		Considering the Chip manufacturer for a *SIM* card as an example build an *IDEF* modeling method.	16

5	What are the various parameters to be considered while establishing computational method? Explain the theoretical units in manufacturing computing.	16
	OR	
6	Elaborate the connotation of *MOT* with respect to the statement quoted below: "Technology economics and business management reflect a relation between the theory and commercial application which is similar to the relation between management and economics".	16
7	Illustrate the relationship among the entities involving sales activities and explain the fundamental characteristics of manufacturing activities.	16
8	Explain in detail the Micro-Nano-Electro Mechanical system. Support your answer with suitable examples.	16

E-Commerce

VI Semester SEE Question Papers

Course codes: 16IM6D5

- Grade Improvement Nov 2020 (16IM6D5)
- Apr/May 2019 (16IM6D5)

16IM6D5

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Grade Improvement Examinations Nov-2020

Industrial Engineering and Management

E-COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Enumerate the merits and demerits of E-Commerce.	02
	1.2	Identify the components of E-Commerce.	02
	1.3	Explain the architecture of E-Commerce.	02
	1.4	Enumerate different applications of E-Commerce.	02
	1.5	List the different models of E-Commerce.	02
	1.6	Summarize the $B2C$ model.	02
	1.7	Find the features of E-Commerce website.	02
	1.8	List the problems faced by online sellers.	02
	1.9	Identify the security issues of E-Commerce.	02
	1.10	What are emerging trends in E-Commerce?	02

PART B

2	a	Analyze transparency issues in $B2B$ models and substantiate your answers.	08
	b	Distinguish between $B2B$ and $B2C$ in terms of serviceability, control and order completion cycle.	08
3	a	Discuss the TCP layer process with a neat block diagram.	08
	b	Differentiate between $IPv4$ and $IPv6$.	08
		OR	
4	a	Segregate with examples between Frames, Forms and Style sheets.	08
	b	Define $HTML$ and why do we need it in E-Commerce applications?	08
5	a	Classify the critical three primary processes which are enhanced in E-business transactions. Give examples.	08
	b	Elucidate with examples, the major different types of E-commerce applications.	08
		OR	

6	a	Assess the impact of $B2B$ markets on the economy of developing countries.	08
	b	How is E-Commerce useful to developing new Entrepreneurs? Substantiate your answers.	08
7	a	"For E-Commerce to be successful, the distribution channel needs to be redesigned"- Comment on this statement with illustrations.	08
	b	How can you use E-Commerce for effective integration of multichannel business? Discuss with suitable examples.	08
8	a	Estimate with reasoning, the trends and the challenges of social commerce.	08
	b	Discern the external drivers from internal drivers for E-Commerce.	08

16IM6D5

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

E - COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Give one example each for horizontal portals and vertical portals.	02
	1.2	What is the purpose of affinity portals?	02
	1.3	Draw basic network architecture for the internet connections.	02
	1.4	Provide general representation of Uniform Resource Locator.	02
	1.5	List different types of e-commerce.	02
	1.6	Give any two examples of more commonly used $B2B$ examples and best practice models.	02
	1.7	What is meant by dynamic pricing?	02
	1.8	"You cannot push a square peg through a round hole". What does this statement means?	02
	1.9	How does the social commerce help the retailers? List any 4 points.	02
	1.10	List the factors affecting environmental and organizational drivers of online selling diffusion (any four factors).	02

PART B

2	a	Web 2.0 characterized especially by the change from static web pages to dynamic or user generated content. How this technology advancement impacted e – commerce? Discuss with a suitable case study.	08
	b	Briefly explain the history of e-commerce by highlighting the chronological events occurred.	08
3	a	With the help of a neat block diagram, explain the basic architecture of internet networking.	08
	b	Explain the web components between the front end and back end activities of a web portal.	08
		OR	
4	a	How does transmission control protocol layer processing work? Use a neat block diagram to explain the mechanism of TCP layer processing.	10
	b	Explain different steps to be followed before a web page is designed to get a good end result.	06

5	a	"The existing legal systems in most developing countries are not sufficient to protect those engaged in e-commerce". Do you agree with above statement? Explain the common legal issues faced by many developing countries.	08
	b	With the suitable examples explain any four models of e-commerce.	08
		OR	
6	a	Briefly outline major players and corresponding requisites involved in a typical e – commerce transaction loop.	10
	b	Elaborate the typical components of e – business model.	06
7	a	With the help of a suitable case study explain the benefits of web enabled channels.	08
	b	What is the importance of dynamic pricing for e – commerce? Explain with the help of an example.	08
8	a	"Social commerce sometimes known to be effective only for impulse purchases". In evident to this statement, explain future of social commerce in a country like India, which has diversified cultural and economic background.	08
	b	Analyze the environmental and organizational drivers of online selling diffusion.	08

Data Mining Technologies

VI Semester SEE Question Papers

Course codes: 16IM6C5 / 12IM7E5, 16IM6C5

- Fast Track Oct 2020 (16IM6C5 / 12IM7E5)
- Apr/May 2019 (16IM6C5)

16IM6C5 / 12IM7E5

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

DATA MINING TECHNOLOGIES (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

			02
1	1.1	Distinguish between 'descriptive' and 'predictive' data mining tasks.	02
	1.2	What is meant by 'topic model' in the context of data mining?	02
	1.3	Mention the three attributes considered for determining the data security.	02
	1.4	Identify the algorithm that uses prior knowledge for mining frequent item sets.	02
	1.5	Mention the two steps involved in data classification.	02
	1.6	Identify a heuristic for selecting the splitting criterion that separates a given data partition.	02
	1.7	What is meant by network planning?	02
	1.8	Mention the appropriate method used for classification to discover structural relationships within noisy data.	02
	1.9	Distinguish between Agglomerative and Divisive hierarchical clustering.	02
	1.10	Categorize the types of attributes that represent the features of a data object.	02

PART B

2	a	Review with suitable examples, several disciplines that strongly influence the development of data mining methods.	08
	b	Examine different approaches to data visualization.	08
3	a	Analyze the major steps involved in data preprocessing.	08
	b	Provide an overview of data reduction strategies.	08
		OR	
4	a	Explain different data smoothing techniques with suitable illustrations.	08
	b	In the real world data, tuples with missing values for some attributes are a common occurrence. Review the methods generally followed to fill in the missing values for the attribute in tuples.	08
5	a	The attribute selection measures provide a ranking for each attribute describing the given training tuples. In the above context, provide an overview of different attribute selection measures that are commonly used for building decision trees.	08

	b	Categorize the metrics for evaluating the performance of 'classifiers'. Also tabulate the measures and the corresponding formulae.	08
		OR	
6	a	Different ensemble methods are employed to improve classification accuracy. Provide a brief note on the same.	08
	b	The class imbalance problem occurs when the main class of interest is represented by only a few tuples. Review the strategies to address this problem.	08
7	a	Illustrate a simple Bayesian belief network for the variable 'Lung Cancer'.	08
	b	Distinguish between Eager Learners and Lazy Learners. Also explain the features of Case Based Reasoning.	08
8	a	Characterize different methods of clustering.	08
	b	Write short notes on the following: i) Visual data mining versus Audio data mining ii) Ubiquitous data mining versus Invisible data mining.	08

16IM6C5

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

DATA MINING TECHNOLOGIES (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Write a note on Data Mining.	02
	1.2	What is data visualization?	02
	1.3	Identify the seminal algorithm for mining frequent item sets in data mining.	02
	1.4	What is meant by Association rule mining?	02
	1.5	What is dimension table?	02
	1.6	Classify different Bayes methods.	02
	1.7	What are Lazy Learners?	02
	1.8	What is the purpose of cluster analysis?	02
	1.9	List different partitioning methods	02
	1.10	Provide an overview on Density based methods.	02

PART B

2	a	Explain any two types of data used for mining.	05
	b	List the applications of data mining: Explain any two in detail	05
	c	What are Data objects & Attribute types? Provide examples.	06
3	a	Explain the four major tasks in Data Preprocessing.	08
	b	Explain the necessity of data integration in data mining.	08
		OR	
4	a	Explain the strategies, “Data transformation” & “Data discretization” in data mining.	08
	b	Briefly explain different methods of mining frequent patterns.	08
5	a	What are basic concepts of classification? Explain the two steps involved in classification.	08
	b	Explain the methodology of decision tree induction in data mining.	08
		OR	

6	a	Explain in detail the two methods of Bayes classification.	08
	b	Explain different methods of Rule based classification.	08
7	a	Explain Bayesian belief networks. Also write a note on training bayesian belief networks.	08
	b	Explain the two methods of classification using frequent patterns.	08
8	a	Explain any two partitioning methods in cluster analysis.	08
	b	Explain any four data mining applications with respect to research frontiers.	08

Financial Management

VI Semester SEE Question Papers

Course codes: 16IM6C4

- Apr/May 2019 (16IM6C4)

16IM6C4

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

FINANCIAL MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of interest factor tables permitted. Tables for factors leading to discounting and compounding conversions permitted.

PART A

1	1.1	Mention any two main areas of decision making as relevant to the area of financial management.	01
	1.2	Write the formula for sinking fund factor with usual notations.	01
	1.3	Distinguish between regular annuity and deferred annuity.	01
	1.4	Mention any one reason as to why the concept of time value of money be considered for all calculations in financial decision making.	01
	1.5	Bring out any significant difference between an ordinary share and a preference share.	01
	1.6	What is the most appropriate measure of risk with respect to a single asset?	01
	1.7	Mention any one of the reasons that make capital budgeting decisions important.	01
	1.8	Write down the governing equation of the Capital Asset Pricing Model($CAPM$).	01
	1.9	Mention any one of the uses of liquidity ratios.	01
	1.10	Write down the equation for any one of the ratios related to capital structure (for leverage).	01
	1.11	Mention any one of the application of profitability ratio for financial analysis.	01
	1.12	Mention any one of the problems related to ratio analysis?	01
	1.13	What is the meaning of the rights issue of equity shares?	01
	1.14	Mention any one advantage of lease financing over hire purchase financing.	01
	1.15	Mention any two methods of venture capital financing.	01
	1.16	Highlight any one significant trend indicating development of venture capital in India.	01
	1.17	Graphically illustrate the cash to cash cycle as applied to working capital assessment.	01
	1.18	Why is inventory levels in the organization included in the working capital calculations?	01
	1.19	What is spontaneous financing in the context of working capital management.	01
	1.20	Distinguish between permanent working capital and temporary working capital.	01

PART B

2 a Discuss the three broad areas of financial decision making. 08

b Explain and illustrate the concept of yield to maturity. 04

c A $Rs.100/-$ per value bond bearing a coupon rate of 12 percent will mature after 7 years. What is the value of the bond if the discount rate is 14 percent? 12 percent? 04

3 a You are considering purchasing the equity stock of MVM company. The current price per share is $.10/-$. You expect the dividend a year hence to be $Rs.100/-$. You expect the price per share of MVM stock a year to have the following probability distribution.

Price a year hence	$Rs.10/-$	$Rs.11/-$	$Rs.12/-$
Probability	0.4	0.4	0.2

i) What is the expected price per share a year hence?

ii) What is the probability distribution of the rate of return on $MVM's$ equity stocks? 08

b A Machine purchased six years back for $Rs.1,50,000$ has been depreciated to a book value of $Rs.90,000/-$. It originally had a projected life of fifteen years and zero salvage value. A new machine will cost $Rs.2,50,000/-$ and result in a reduced operating cost of $Rs.30,000/-$ per year for the next nine years. The older machine can be sold for $Rs.50,000/-$. The machine also will be depreciated on a straight line method on nine year life with salvage value of $Rs.25,000/-$. The company's tax rate is 50% and cost of capital is 10%. Determine whether the old machine should be replaced. 08

OR

4 a Vardhman Limited's earnings and dividends have been growing at a rate of 18 percent per annum. This growth is expected to continue for 4 years . After that the growth rate will fall to 12 percent for the next 4 years. Thereafter the growth rate is expected to be 6 percent forever. If the last dividend per share was $Rs.2.00$ and the investors required rate of return on Vardhaman's equity is 15 percent, what is the intrinsic value per share? 08

b The cash flows associated with three projects P, Q and R are given below.

Year	*Net cash flow*		
	P	Q	R
0	(2000)	(2000)	(2000)
1	1,400	500	500
2	600	1,100	500
3	400	900	1600

Calculate the net present value of each project at discount rate of 0 percent, 5 percent, 10 percent, 15 percent, 25 percent and 30 percent. Plot the results on a graph paper and interpret. 08

5 a Explain the important profit margin ratios. 04

b Determine the sales (i.e revenues from operations) of a firm with the following financial data.

Current ratio	1:5
Acid test ratio	1:2
Current liabilities	8,00,000
Inventory turnover ratio	5 *times*

08

	c	What are percentage format financial statements? How are they useful?	04
		OR	
6	a	What is the importance of ratio analysis? Briefly discuss the importance of the following accounting ratios. i) Liquidity ratio ii) Debt equity ratio iii) Stock turnover rate iv) Ratio of debtors to turnover.	08
	b	Complete the balance sheet and sales data (fill in the blanks) of a firm with the following financial data. Debt/equity ratio= 0.6 Acid test ratio= 1.2 Total assets to turnover ratio = 1.5 Day's sales outstanding recievables in trade= $40 days$ cost of goods sold as a percentage of total revenue 80 percent inventory turnover ratio = 5 Balance sheet	

Equity capital	50,000
Retained earnings	60,000
Plant and equipment	– – – –
Inventories	– – – –
Trade recievables	– – – –
Cash and cash equivalents	– – – –
Revenue from operations	– – – –
other income	*NIL*

08

7	a	Define the following terms in relation to an equity share: Par value, issue price, book value and market value.	04
	b	What are the features of venture capital?	04
	c	The Hypothetical Manufacturers Ltd (HML) has taken a plant on lease, valued at $Rs.20$ crores. The lease arrangement is in the form of leveraged lease. HML is the equity participant and the Hypothetical Bank Limited (HBL) is the loan participant. They find the investment in the ratio of 2:8. The loan from HBL carries a fixed rate of interest of 19 percent, payable in 6 equated installments. The lease term is 6 years with lease rental payable annually in arrear. i) Compute the equated annual installment from the point of view of HBL. ii) If the lease rate is unknown and $HBL's$ pre-tax yield is 25 percent, what is the minimum lease rate that must be quoted?	08
8	a	The following annual figures relate to XYZ Co.	

Sales (at two month's credit)	*Rs* 3600,000/–
Materials consumed (suppliers extend two months credit)	*Rs* 9,00,000/–
Wages paid (Monthly in arrears)	*Rs* 7,20,000/–
Manufacturing expenses outstanding at the end of the year (cash expenses are paid month in arrear)	*Rs* 80,000/–
Total administrative expenses paid as above	2,40,000/–
Sales promotion expenses, paid quarterly in advance	1,20,000/–

	The company sells its products on a gross profit of 25% consisting depreciation as part of the cost of production. It keeps one month's stock each of raw materials and finished goods and a cash balance of $Rs.1{,}00{,}000/-$. Assuming 20% safety margin, work out the working capital requirements of the company on 'cash cost basis'. Ignore work in process.	10
b	Discuss the important forms of working capital advance given by banks. What is the meaning of an letter of credit arrangement?	06

Advanced Operations Research

VI Semester SEE Question Papers

Course codes: 12IM6D3

- Apr/May 2018 (12IM6D3)
- Apr/May 2017 (12IM6D3)
- Apr/May 2016 (12IM6D3)

12IM6D3

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

ADVANCED OPERATIONS RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

7 Answer FIVE full questions from Part B.

PART A

1	1.1	In the two phase method, an _____ variable is never considered for re-entry into the basis.	01
	1.2	If in the integer *LP* problem, few of the basic variables do not possess non-negative integer value, then an additional constraint known as _____.	01
	1.3	Goal programming is used for ballistic _____ management.	01
	1.4	In Goal Programming, a goal constraint having over achievement variable is expressed as a _____ constraint.	01
	1.5	Resource leveling is the process of _____ the utilization of resources in a project.	01
	1.6	Crashing is the process of reducing the total time that it takes to complete a project by expending _____.	01
	1.7	The mean of exponential distribution is the _____ time between arrivals.	01
	1.8	the payment of telephone bills by cheque or cash is the example of _____ discipline.	01
	1.9	In Markov analysis, state probabilities must _____ to one.	01
	1.10	A zero element in the transition matrix indicates that the transition is _____.	01
	1.11	Optimally is indicated for a maximization problem when all elements in the $c_j - z_j$ rows are _____, while for minimization problem all elements must be _____.	02
	1.12	In integer programming, each fractional cut that is made, requires the addition of a _____ or _____ constraint.	02
	1.13	Network is the graphical display of a project that contains both _____ and _____.	02
	1.14	The dynamic arrival process is controlled by _____ facility and the _____.	02
	1.15	A problem is classified as Markov chain provided states are collectively _____ and _____ exclusive.	02

PART B

2	a	Define slack and surplus variables in a linear programming problem.	04
	b	A company sells two types of fertilizers, one is liquid and the other is dry. The liquid fertilizer contains 2 units of chemical A and 4 units of chemical B per jar the dry fertilizer contains 3 units of each of the chemicals A and B per carton. The liquid fertilizer sells $Rs.3$ per jar and dry fertilizer sells for $Rs.4$ per carton. A farmer requires at least 90 units of chemical A and 120 units of chemical B for his farm. How many of each fertilizer should the farmer purchase to minimize the cost while meeting the requirements. Solve by using Two Phase Simplex Method.	12
		OR	
3		Solve the following integer linear programming problem by Branch and Bound method. Max $Z = 6x_1 + 8x_2$ subject to $x_1 + 4x_2 \leq 8$ $7x_1 + 2x_2 \leq 14$ where x_1, x_2 are non-negative integers.	16
4	a	Why all goal linear programming problems are minimization problem? Give reasons.	06
	b	The elastic firm has two products coming on the market: Zigs and Zags. To make a Zig, the firm needs 10 units of product A and 15 units of product B. To make a Zag, they need 20 units of product A and 15 units of product B. there are only 2,000 units of product A and 3200 units of product B available to the firm. The profit on a Zig is $Rs.4$ and on a Zig it is $Rs.6$. Management objectives in the order of their priority are: i) Produce exactly 50 Zigs ii) Achieve a target profit of at least $Rs.750$ iii) Use all of the product B available. Formulate this as a goal programming problem.	10
		OR	
5	a	Explain Goal Programming applications with respect to the following areas: i) Marketing ii) Finance iii) Production iv) Human Resource Development.	08
	b	Use the Wolfe's method to solve the quadratic programming problem: Max $Z = 4x_1 + 6x_2 - 2x_1^2 - 2x_1x_2 - 2x_2^2$ subject to $x_1 + 2x_2 \leq 2$ and $x_1, x_2 \geq 0$	08
6	a	Discuss the general algorithm used in solving problem of dynamic programming approaches.	10
	b	Regardless of the type or size, following terms are common in dynamic programming problem: discuss them: i) Stage ii) State iii) Return Function.	06

OR

7 The following table gives data on normal time, cost and crash time and cost for a project:

Activity	Normal		Crash	
	Time (weeks)	Cost (Rs)	Time (weeks)	Cost (Rs)
1 – 2	3	300	2	400
2 – 3	3	20	3	30
2 – 4	7	420	5	580
2 – 5	9	720	7	810
3 – 5	5	250	4	300
4 – 5	0	0	0	0
5 – 6	6	320	4	410
6 – 7	4	400	3	470
6 – 8	13	780	10	900
7 – 8	10	1000	9	1200

Indirect cost is $Rs.50$ per week.

a) Draw the network and identify the critical path with double line
b) What are the normal project duration and associated cost?
c) Find out the total float associated with each activity
d) Crash the relevant activities symmetrically determine the optimal project completion time and cost. 16

8 a Explain the following features of generalized queuing system:
i) Length of the queue
ii) System
iii) Waiting time
iv) Total time. 08

b Arrivals at a telephone booth are considered to be Poisson with an average time of 10 minutes between one arrival and the next. The length of phone calls is assumed to be distributed exponentially with mean 3 minutes.
i) What is the probability that a person arriving at the booth will have to wait?
ii) What is the average length of the queues that forms from time to time?
iii) The telephone department will install a second booth when convinced that an arrival would expect to have to least three minutes for the phone. By how much the flow of arrivals is to be increased in order to justify a second booth? 08

OR

9 a Discuss the following with respect to a customer's behavior in a queue:
i) Balking
ii) Collusion
iii) Jockeying. 06

b A person repairing radios finds that the time spent on radio sets has exponential distribution with mean 20 minutes. If the radios are repaired in the order in which they come in and their arrival is approximately Poisson with an average rate of 15 for 8 hour day, what is the repairman's expected idle time each day? How many jobs are ahead of the average set just bought in? 10

10	a	State and discuss the characteristics of a Markov Chain.	10
	b	State the steps for constructing Matrix of Transition Probabilities of Markov Chain.	06
		OR	
11	a	State the application of Markov analysis.	04
	b	The school of international studies for population find out, though its survey, that the mobility of the population (in percent) of a state to a village, town and city is in the following percentages.	

		TO		
		$Village$	$Town$	$City$
$From$	$Village$	40	30	30
	$Town$	20	60	20
	$City$	20	40	40

What will he the proportion of population in village, town and city after two years, given that the percent population has proportions of $0.6, 0.3$ and 0.1 in village, town and city respectively? 12

12IM6D3

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

ADVANCED OPERATIONS RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

8 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

9 Answer FIVE full questions from Part B.

PART A

In this part, there are statements made based on the concepts related to this course. You are required to provide one line of reasoning to substantiate theses statements (why? Part of understanding these concepts is required to be articulated) and its relevance to problem solving operation's research modeling frameworks. (you may provide case situations, diagrams, illustrations to reason out as necessary)

1	1.1	The revised simplex method substantially reduces the computational effort as compared to the Dantzig's simplex method.	01
	1.2	Shadow prices with respect to resource constraints have their utility in evaluating the resource priorities and economic significance in managing the resources.	01
	1.3	The two phase simplex method diagnoses the condition of infeasibility at the end of Phase I and hence is efficient compared to the regular simplex method.	01
	1.4	The branch and bound algorithm is the most widely used method for solving both pure and mixed integer programming problems in practice. Most commercially available computer codes for solving integer programs are based on this approach.	01
	1.5	In goal programming, all the objectives are assigned target levels for achievement and a relative priority on achieving these levels. Goal programming treats these targets as goals to aspire for and not as absolute constraints.	01
	1.6	Inmost engineering problems, -especially real world formulations- the objective is to optimize (maximize or minimize) a criterion (or objective) function subject to several constraints. These could be linear or non linear in nature.	01
	1.7	The mathematical technique of Lagrange multipliers has been developed to convert constrained optimization problems into unconstrained optimization problems.	01
	1.8	The optimal set of decisions in a multistage decision process problem has the property that whatever the initial stage, state and decisions are, the remaining decisions must constitute an optimal sequence of decisions for the remaining problem, with stage and state resulting from the first decisions (or occurring naturally) considered as initial conditions.	01

1.9	In dynamic programming, one solves multivariable optimization problems sequentially, or one stage at a time. Hence it will be necessary to keep track of all the returns accumulated in our decision process as we proceed from stage to stage.	01
1.10	If the search for the optimal solution of a given LPP(Linear Programming Problem) starts with a dual feasible basis (A canonical tableau where the optimality conditions are satisfied), and search for a primal feasible basis by moving from one dual feasible tableau to another dual feasible tableau . the approach is called the dual simplex method.	01
1.11	When the process is irreducible, recurrent, positive and a periodic the Markov chain modeling is applied the resultant Markov chain is ergodic.	01
1.12	If you are modeling a process that does not terminate after some length of time, the model will take the form of so called absorbing Markov chain.	01
1.13	Given a transition matrix, we can construct a unique transition diagram; Given a transition diagram we can construct a unique transition matrix.	01
1.14	In the context of queuing models, a customer is anything that arrives occupies a server for some period of time, and departs. It need not be a human, or for that matter a physical object. It could be a job, an order a transaction, a message or some similar abstraction.	01
1.15	The relationship linking L(Mean length of system), W(Mean waiting time in the system) and R (mean throughput rate) is expressed as simple direct relationship $L = RW$ (little's formula). This formula applies on a very broad scale to all kinds of problems. (say for conveyor assembly line for example).	01
1.16	The simplest arrival process to deal with is a Poisson process, in which the times between arrivals are independent negatively exponentially distributed random variables. The same process viewed another way gives the random variables for the number of customers arriving in any interval (hence the name for the process).	01
1.17	One of the most efficient algorithms for solving shortest route problems is known as the Dijkstra's algorithm.	01
1.18	The time required to complete the various activities in a research and development project is generally not known a priori. Thus $PERT$ incorporates uncertainties in activity times in the analysis.	01
1.19	For large projects the approach to solving CPM problems deployed is the mathematical modeling methods. The basic among them is the LPP models.	01
1.20	The basic assumption in CPM models is that the activity times are proportional to the level of resources allocated to them. By assigning the additional resources (Capital, People, Materials and Machines) to an activity, its duration can be reduced to a certain extent.	01

PART B

2	a	A factory manufactures three products. Three resources- technical services, labor and administration are required to produce these products. The following table gives the requirements on each of the resources for the three products.	

$Product$	$Resources(hours)$			$Unit\ Profit$ (\$)
	$Technical\ service$	$Labor$	$Administration$	
1	1	10	2	10
2	1	4	2	6
3	1	5	6	4

There are 100 hours of technical services available, 600 hours of labor, and 300 hours of administration , In order to determine the optimal produce mix which will maximize the total profit, the following linear program was solved:

Maximize $Z = 10x_1 + 6x_2 + 4x_3$

Standards to $x_1 + x_2 + x_3 \leq 100$ $(technical)$

$10x_1 + 4x_2 + 5x_3 \leq 600$ $(labor)$

$2x_1 + 2x_2 + 6x_3 \leq 300$ (administration)

$x_1, x_2, x_3 \geq 0.$

Where $x_1, x_2,$ and x_3 are the number of product 1, product 2 and product 3 produced. The optimal solution is given by the following tableau, where $x_4, x_5,$ and x_6 are the slack variables:

C_j	10	6	4	0	0	0	$Constants$
$Basis$	x_1	x_2	x_3	x_4	x_5	x_6	
x_2	0	1	5/6	10/6	−1/6	0	400/6
x_1	1	0	1/6	−4/6	1/6	0	200/6
x_6	0	0	4	−2	0	1	100
$C\ Row$	0	0	−16/6	−20/6	−4/6	0	$Z = 4400/6$

Using sensitivity analysis, answer the following with respect to the above optimal tableau:

i) What should be the profit of product 3 before it becomes worthwhile to manufacture? Find the most profitable product mix if the profit on product 3 were increased to \$50/6.

ii) What is the range on the profit of product 1 so that the current solution is still optimal?

iii) It is believed that the estimates of the available hours of technical services might be wrong. The correct estimate is $10 + 10\lambda$ where λ is some unknown parameter. Find the range of values λ within which the given product mix is still optimal.

iv) Determine the shadow prices of all the resources. 16

OR

3 a Use Two-Phase simplex method to

Maximize, $Z = 2x_1 + x_2 + x_3$

Subject to $4x_1 + 6x_2 + 3x_3 \leq 8$

$3x_1 - 6x_2 - 4x_3 \leq 1$

$2x_1 + 3x_2 - 5x_3 \geq 4$

$x_1, x_2, x_3 \geq 0.$ 08

b Express this LPP in the format of revised Simplex method. Carry out first two iterations using the revised simplex method.

Maximize, $Z = x_1 + 5x_2 + 3x_3$

Subject to $x_1 + 2x_2 + x_3 = 3$

$2x_1 - x_2 = 4$

$x_1, x_2, x_3 \geq 0$ 08

4 a Camyo Manufacturing produces four parts that require the use of a lathe and a drill press. The two machines operate 10 hours a day. The following table provides the time in minutes required by each part:

7 The following tables gives the activities in a construction project and other relevant information:

Activity	Description	Immediate Predecessor	Normal		Crash		Cost slope	Rank
			Time	cost	Time	cost		
A	Dismantle and inspect	-	2	100	1	140	40	II
B	Subcontract repairs	A	6	200	2	500	75	IV
C	Buy spares	A	3	100	1	140	20	I
D	Rebuilt /repair	A	4	500	2	800	150	V
E	Assemble	B, C, D	2	200	1	250	50	III
F	Trial run	E	1	50	Not possible	250	-	-

Determine the least cost schedule for the project. 16

8 a The queue discipline is a parameter that explains how the customers arrive at a service facility. The commonly referred structures of queue disciplines are

i) First in first out ($FIFO$)
ii) First in last out($FILO$)
iii) Served in random order ($SIRO$)
iv) Priority scheduling
v) Processor (or Time) sharing.

Give one practical example for each one of the disciplines mentioned above. 08

b With usual Notation and based on the notion of birth Death Processes. Bring out the relationships between L, W, Lq and Wq: for a typical $M/M/1$ queuing system. 08

OR

9 a A blood donation drive organized in a small college operates from a room with 2 beds, and on an average it takes each donor 20 minutes to enter the room, finish the screenings, donate blood and leave. There is enough space to seat any number of waiting students outside the room. The students arrive according to a Poisson process with a mean inter arrival time of 25 minutes. What is the probability of an arriving student having to wait to enter the room?

What is the utilization of the beds in the room? 10

b How would you use the Kendall & Lee's notation to describe the characteristics of various types of queuing system? Give an example. 06

10 a ABC hospital operated on a charity basis. All expenses are paid by the government. Recently the board of governors of the hospital has been complaining about the size of the budget and is in insisting that the hospital cut expenses. The major area of concern has been the cost of keeping the patients in intensive care. This cost has averaged $Rs.1000$ per week per person compared to only $Rs.500$ per week per person for keeping patients in the ICU at the beginning of the week, 50% will be there at the end of the week and 50% will be moved to a ward. Of the patients in the ward at the beginning of the week, 50% will be there at the end of the week, 10% will get worse and be transferred to ICU: and 40% will become outpatients.

Of those persons who are outpatients at he beginning of the week; 85% will remain outpatients at the end of the week; 10% will be admitted to a ward and 5% will be admitted to ICU. The board of the government believes the criteria for keeping patients in ICU are strict and has instructed the ICU staff to relax the criteria so that only 40% of the ICU patients remain in ICU each week and 60% are transferred to ward. The staff insists that if this is done, 20% of the ward patients will be going into ICU each week and only 30% will be transferred to out patients status. The percentage ward patient remaining will be still be 50%. There will be no change in outpatient status. Will the policy advocated by the board of governors actually save money? 12

b List out any four important areas of applications of the Markov Chain. Modeling approach. 04

OR

11 a A market survey has been conducted to determine the movements of people between types of residences. Two hundred apartment dwellers were asked if their previous residence was an apartment, a townhouse, their own house, or a rented home. Similarly, 200 townhouse dwellers were asked about their previous residences, and so on. The results of the survey are tabulated below:

Present residence	*Previous residence*				
	Apt.	*Townhouse*	*Own house*	*Rented House*	*Total*
Apartments	100	20	40	40	200
Townhouse	150	40	0	10	200
Own house	50	20	120	10	200
Rented house	100	20	20	60	200
Total	400	100	180	120	800

The data are believed to be representative of the behavior of the population at large. Develop a model and answer the following:

i) What is the probability that someone who is now is an apartment will own his own home after two moves?
ii) What is the probability that the same person will own his own home after two moves but not sooner?
iii) How many moves will occur on the average, before a homeowner will live in a townhouse?
iv) What, if anything can you say about the long term demand for townhouses, relative to the other residence types?
v) Consider the validity of your assumption. 12

b Discuss the property of ergodicity and its significance to modeling using Markov chain structures. 04

12IM6D3

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

ADVANCED OPERATIONS RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

4 Answer FIVE full questions from Part B.

PART-A

1	1.1	The advantage of the two phase simplex method over the regular simplex method is that the _____ condition in the solution to a given LPP is identified quickly (earlier) during the process of solving the LPP.	01
	1.2	The revised simplex method requires the expression of the given LPP in the _____ structure to initiate the process of solving using the matrix inversion technique to change the basis at every iteration.	01
	1.3	The integer programming formulation which requires that all decision variables have necessarily to be integers is known as the _____ integer program.	01
	1.4	The shadow price is defined as the _____ worth of one additional unit of the resources expressed as constraints in a given LPP model.	01
	1.5	The branch and bound technique of solving a given integer programming problem requires the solution of many _____ to the original problem to generate the branches and bound the optimal solution.	01
	1.6	Goal programming formulation requires the specification of decision variables and _____ with respect to each of the goals to be met in a given problem situation.	01
	1.7	The goal programming formulation involves the objective function which involves the _____ of the sum of deviational variables.	01
	1.8	The Kuhn Tucker conditions for the non linear programming problem can be said to be the necessary and _____ conditions for the optimal solution of the non linear program.	01
	1.9	The general structure of a quadratic programming problem involves the non linear objective function and _____ constraints.	01
	1.10	If the constraint of a quadratic programming problem is $\sum_{j=1}^{n} a_{ij}x_j \leq b_j$ and the non negativity restraints are $x_j \geq 0$. The slack variables _____ are introduced into the constraints.	01
	1.11	The dynamic programming problem involves the division of the problem into _____ and define the state transition function to link the various stages of the solution methodology.	01
	1.12	The principles of optimality which form the basis for solving a given problem using dynamic problem formulation is known as the _____ principle of optimality.	01

	1.13	The schedule of a project network which drives the total cost of the project inclusive of the direct cost and indirect cost to the absolute minimum is known as the _____ schedule.	01
	1.14	One of the most efficient algorithms to determine the solution for the shortest route problem is known as the _____ algorithm.	01
	1.15	The _____ property which is one of the basis for realizing the continuous time process for modeling the queuing process, helps in assuming the exponential distribution for time between arrivals.	01
	1.16	Nearly all simple queuing models (as well as a number of non queuing models) are special cases of the _____ process which is itself a special case of the general continuous time Markov process.	01
	1.17	If for all n $p\{X_n = j_n/X_{n-1} = j_{n-1}, X_{n-2} = j_{n-2} - - - - - - X_0 = j_0\} = p\{X_n = j_n/X_{n-1} = j_{n-1}\}$ the process is said to be discrete state, discrete parameter Markov process or more generally known as _____.	01
	1.18	The property of _____ is exhibited by a discrete p-state, discrete parameter process, when the process is irreducible, recurrent, positive and aperiodic.	01
	1.19	If you are modeling a process that does terminate after some length of time the model will usually take the form of a so called _____ Markov chain.	01
	1.20	$p^{(2)}$ is called _____ transition matrix; its elements, two –step transition probabilities. They give conditional probabilities for states at time 2 under varying possible conditions for the state at time zero.	01

PART-B

2	a	Solve the following LPP use the two phase method Minimize $Z = 4x_1 + x_2$ Subject to the constraints: $3x_1 + x_2 = 3$ $4x_1 + 3x_2 \geq 6$ $x_1 + 2x_2 \leq 4$ $x_1 x_2 \geq 0$	08
	b	Use revised simplex method to solve the following LPP Minimize $Z = x_1 + 2x_2$ Subject to $2x_1 + 5x_2 \geq 6$ $x_1 + x_2 \geq 2$ $x_1, x_2 \geq 0$	08
		OR	
3	a	Consider a diet problem in which a college student is interested in finding a minimum cost diet that provides at least 21 units of vitamin A and 12 units of vitamin B from five foods of the following properties:	

$Food$	1	2	3	4	5
$Vitamin\ A\ content$	1	0	1	1	2
$Vitamin\ B\ content$	0	1	2	1	1
$Cost\ per\ unit$	20	20	31	11	12

The corresponding linear program is
Minimize $Z = 20x_1 + 20x_2 + 31x_3 + 11x_4 + 12x_5$
Subject to
$x_1 + x_3 + x_4 + 2x_5 \geq 21$ (Vitamin A constraint)
$x_2 + 2x_3 + x_4 + x_5 \geq 12$ (Vitamin B constraint)
Where x_i- amount of food 'j' bought for $j = 1, 2, - - -5$. Solution obtained for the LP using the computer program is given below:
Optimal solution $x_1 = x_2 = x_3 = 0, x_4 = 3, x_5 = 9$
Optimal value Min $Z = 141$
Shadow prices 1.0 and 10.0 for constraints 1 and 2 respectively
Opportunity costs $19.0, 10.0, 10.0$ and 0.0 for variables x_1, x_2, x_3, x_4 and x_5 respectively.

Ranges on objective function coefficients

Variable	*Lower Limit*	*Present Value*	*Upper Limit*
x_1	1	20	∞
x_2	10	20	∞
x_3	21	31	∞
x_4	6	11	12
x_5	11	12	22

Ranges on RHS constants

Row	*Lower Limit*	*Present Value*	*Upper Limit*
1	12	21	24
2	10.5	12	21

Using the above information, answer the following:

i) For what range of values of $C3$ (cost of food 3) is the current solution still optimal?

ii) Determine the optimal solution is the minimum diet cost of food 1 is reduced to 5 cents per unit.

iii) What is the optimal solution and the minimum diet cost if the unit cost of food 4 is reduced to 7 cents.

iv) A student's favorite is food 2 and she would like to include 2 units of it in the diet. Explain how this will affect the minimum cost diet.

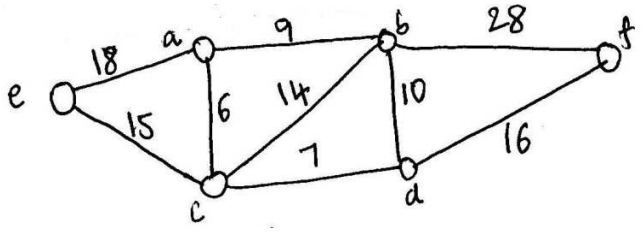
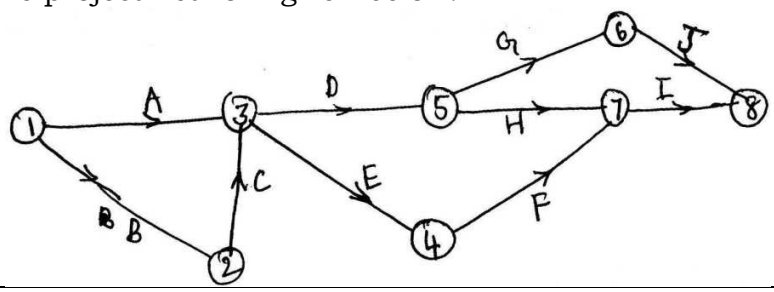
v) Suppose the vitamin A requirement is increased from 12 to 15 units. A local pharmacist offers the additional 3 units of vitamin B at a cost of 12 cents per unit. Would you accept the offer? Why or why not? 12

b List the steps of the Gomory's cutting plane method for solving an integer programming problem. 04

4 a Camyo manufacturing produces four parts that require the use of lathe and a drill press. The two machines operate 10 hours a day. The following table provides the time in minutes required by each part

Production time in minutes

Part	*Lathe*	*Drill Press*
1	5	3
2	6	2
3	4	6
4	7	4

		It is derives to balance the two machines by limiting the differences between their total operations to at most 30 minutes. The market demand for each part is atleast 10 units. Additionally the number of units of part 1 may not exceed that of part 2. Formulate the problem as a goal programming model.	08
	b	Use Wolfe's method to solve the quadratic programming problem $f(x) = x_1^2 + x_2^2 - 2x_1 - 4x_2$ Subject to $x_1 + 4x_2 \leq 5$ $2x_1 + 3x_2 \leq 6$ $x_1, x_2 \geq 0$	08
		OR	
5	a	Solve the following problem using only the Kuhn-Tucker conditions $f(x) = 100 - 1.2x_1 - 1.5x_2 + 0.3x_1^2 + 0.05x_2^2$ Subject to: $g_1(x) = x_1 + x_2 \geq 35$ $g_2(x) = x_1 \geq 0$ $g_3(x) = x_2 \geq 0$	08
	b	A company has two machines that can manufacture a product. Machine 1 makes 2 units per hour while machine 2 makes 3 per hour. The company has an order for 160 units. Each machine can be run for 40 hours on regular time; however overtime is available. The company's goal in order of importance are to: i) Avoid falling short of meeting the demand. ii) Avoid any overtime of machine 2 beyond 10 hours. iii) Minimize the sum of overtime (assign differential weights according to the relative costs of overtime hours, assuming that the operating cost of the two machines are the same). iv) Avoid under-utilisation of regular working hours (assign weights according to the productivity of the machines).	08
6	a	List the main features of a dynamic programming model. Briefly bring out its relevance.	04
	b	Apply Dijkastra's algorithm to find the shortest route from s to t in the graph given below:  	12
		OR	
7	a	Consider the project network given below:  	

The data for normal times, crash times and crashing costs are given as follows:

Job	$Normal\ Time\ (days)$	$Crash\ Time\ (days)$	$Cost\ of\ crashing\ per\ day$ (\$)
A	10	7	4
B	5	4	2
C	3	2	2
D	4	3	3
E	5	3	3
F	6	2	5
G	5	4	1
H	6	4	4
I	6	3	3
J	4	3	3

Let T represent the earliest completion time of the project

i) Determine the maximum and minimum value of T

ii) Given that the overhead costs as \$5 per day, determine the optimal duration of the project in terms of both the crashing and overhead costs by direct enumeration method. 14

b Mention any two applications of the minimal spanning tree problem with brief of the problem and suitability of the minimal spanning tree algorithm. 02

8 a What is a Markovian queuing model? 03

b Explain your understanding of the relationship between the arrival rate λ and the average inter arrival time. What are the units describing each variable? 03

c In a bank operation, the arrival rate is 2 customers per minute. Determine the following:

i) The average number of arrivals during 5 minutes.

ii) The probability that no arrivals will occur during the next 5 minutes.

iii) The probability that at least one arrival will occur the next 5 minutes.

iv) The probability that the time between successive arrivals is at least 3 minutes. 10

OR

9 a Describe Kendalls and lee's notation for identifying queuing models. Give two examples to clarify the notation 04

b What is Little's formula and what is its use? 04

c Consider an airport with two runways, one of which is used solely for take off's and other solely for landings. Assume that a plane, whether landing or take off will occupy a runway for an average of 2 minutes where "by occupy" we mean "prevent use by any other plane". Delay on the ground although unpleasant for passengers, does not pose a safety hazard; delay on the other hand is of serious concern. Suppose that the regulation authority specify that the mean delay in the air (i.e w_q) must not exceed 10 minutes. Assume that the planes arrive according to a poisson process and construct a model from the information given above to answer the following questions:

		i) What is the maximum tolerable load on the airport in terms of the mean number of planes that can arrive per hour? ii) Would it make much difference if you were told that the standard deviation of the time a plane will occupy a run way is 1 minute? iii) What practical suggestion can you make as to what might be done to increase the load capacity of the airport, short of building new runways?	08
10	a	Explain how to construct a n step transition probabilities of a typical Markov chain. Give an example to illustrate the construction of TPM.	04
	b	A manufacturing company has a certain piece of equipment that is inspected at the end of each day and classified as just overhauled, good, fair or inoperative. If the item is inoperative it is overhauled, a procedure that takes one day. Let us denote the classification as states $1, 2, 3$ and 4 respectively. Assume that the working conditions of the equipment follows a Markov chain with the following transition matrix Tomorrow $$P = Today \begin{matrix} \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{matrix} \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{bmatrix} 0 & 3/4 & 1/4 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$ If it costs Rs $125/-$ to overhaul a machine (including cost time) on the average and Rs $75/-$ as production lost if a machine is found inoperative. Using the steady – state probabilities. Compute the expected cost per day for maintenance of the piece of equipment.	12
		OR	
11	a	Define the following terms (Give illustrations wherever necessary) i) Absorbing states ii) Ergodic Markov chain iii) Irreducible Markov chain iv) Chapman- kolmogrov equation	06
	b	Suppose that new razor blades were introduced in the market by three companies at the same time. When they introduced each company had an equal share of the number, but during the first year the following changes took place: i) Company A retained 90 percent of its customers, lost 3 percent to B and 7 percent to C. ii) Company B retained 70 percent of its customers, lost 10 percent to A and 20 percent to C. iii) Company C retained 80 percent of its customers, lost 10 percent to A and 10 percent to B. Assuming that the buying habits of the customer remains same. What are the market shares of the three companies at the end of first year? The second year? What are the long-run market shares of the three companies?	10

Applied Statistics

VI Semester SEE Question Papers

Course codes: 12IM6C3

- Apr/May 2018 (12IM6C3)
- Apr/May 2017 (12IM6C3)
- Fast Track Jul 2016 (12IM6C3)
- Apr/May 2016 (12IM6C3)

12IM6C3

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

APPLIED STATISTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

4 Answer FIVE full questions from Part B.

5 Use of statistical tables permitted.

PART A

1	1.1	if $P(x) = \begin{cases} \frac{x}{15} & x = 1,2,3,4,5 \\ \theta & elsewhere \end{cases}$ then, $P(x = 0 \; or \; x = 2) =$ _____.	01
	1.2	Thye moment generating function of the random variable whose moments are $\mu_r^{'} = (r+1)!\, 2^r$	02
	1.3	In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, the probability that a random sample of 5 pages will contain no error _____.	02
	1.4	X is a Gamma distribution with parameter $\lambda = 2$. Mean and variance are related as _____.	01
	1.5	The joint probability density function of a two dimensional random variable (x, y) is given by $f(x,y) = \begin{cases} 2, & \phi < x < 1, \phi < y < x \\ \phi, & otherwise \end{cases}$. The marginal density of x is _____.	01
	1.6	The test statistic used in a test for proposition of defectives based on *'d'* defectives observed in parts is _____.	02
	1.7	Type – II error is defined as _____.	01
	1.8	What is the difference between contingency table test and goodness of fit test?	02
	1.9	In testing for equality of two means the pooled variance (i.e., unbiased estimator of the common population variance σ^2) is given by _____.	01
	1.10	The expression for determining sample size while estimating mean is _____.	01
	1.11	Which hypothesis is tested by Wilcoxon Rank sum test?	01
	1.12	Least squares estimate for β in matrix form is _____.	01
	1.13	What are the hypothesis statements in a test of regression with 3 independent variables?	02
	1.14	What is coefficient of multiple determination?	02

PART B

2	a	A perfect coin is tossed twice. Find the moment generating function of the number of heads. Hence find the mean and variance.	08
	b	Derive moment generating function of $f(x) = \theta e^{-\theta x}, x \geq \theta$. Find the density function, mean and variance.	08
		OR	
3	a	The diameter of an electric cable X is continuous random variable with density $$f(x) = \begin{cases} kx(-1x), & if\ \theta \leq x < 1 \\ \theta, & elsewhere \end{cases}$$ Find: i) k ii) $F(x)$ iii) $p\left(x < \frac{1}{2} \middle| \frac{1}{3} < x < \frac{2}{3}\right)$.	08
	b	The distribution function of a random variable X is given by $F(x) = 1 - (1 + x)e^{-x}, x \geq \theta$. Find the density function, mean and variance.	08
4	a	In a component manufacturing industry, there is a small probability of $\frac{1}{500}$ for any component to be defective. The components are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing: i) Atleast one defective and ii) At most one defective in a consignment of 1000 packets.	08
	b	Suppose that the life time of a certain kind of an emergency backup battery (in hours) is a random variable X having the Weibull distribution with $\alpha = 0.01$ and $\beta = 0.5$. Find the probability that such a battery will: i) Last more than 300 hours ii) Not last 100 hours Also find the mean life time of these batteries.	08
		OR	
5	a	Suppose that the reaction time in seconds of a person can be modeled by a log normal distribution with parameter values $\mu = -0.35$ and $\sigma = 0.2$. i) Find the probability that the reaction time is less than $0.6s$ ii) Find the reaction time that is exceeded by 95% of the population.	08
	b	A case of wine has 12 bottles, 3 of which contain spoiled wine. A sample of 4 bottles is randomly selected from the case. i) Find the probability distribution for x, the number of bottles spoiled wine in the sample ii) What is the probability that at least one bottle is spoiled? iii) Obtain the mean and variance of x.	08
6	a	If the joint density function of the two random variables X and Y be $$f(x,y) = \begin{cases} e^{-(x+y)} & x \geq 0, y \geq 0 \\ 0 & otherwise \end{cases}$$ Find: i) $P(x < 1)$ ii) $P(x + y < 1)$	08

	b	Let the random variable X denote the time until a computer server connects to your machine (in milli seconds) and let Y denote the time until the server authorizes you as a valid user (in milli seconds). Each of these random variables measures the wait from a common starting time and $x < y$. The joint density function of X and Y is $$f_{XY}(x,y) = 6 \times 10^{-6} e^{-0.001x-0.002y} \; for \; x < y$$ i) Obtain marginal distribution of X and Y ii) Find $P(x \leq 1000, Y \leq 2000), P(Y > 2000)$	08
		OR	
7	a	Let X be a Bernoulli random variable with probability mass function $$f(x,p) = p^x(1-p)^{1-x}, x = 0,1$$ where p is the parameter to be estimated. Obtain the maximum likelihood estimator of p.	08
	b	i) An electronic company manufactures resistors that have mean resistance of 100Ω and a standard deviation of 10Ω. The distribution of resistance is normal. Find the probability that a random sample of $n = 25$ resistors will have an eaverage resistance of fewer than 95Ω. ii) An article in the journal Materials Engineering describes the results of tensile adhesion tests on 22 $U-700$ alloy specimens. The load at specimen failure is as follows (in MPa)	

19.8	10.1	14.9	7.5	15.4	15.4	15.4	18.5	7.9	12.7	11.9
11.4	11.4	14.1	17.6	16.7	15.8	19.5	8.8	13.6	11.9	11.4

	Construct 95% confidence interval for μ	08
8 a	Consider the following frequency table of observations on the random variable X.	

$Values$:	0	1	2	3	4
$Observed\ Frequency$:	24	30	31	11	4

	i) Based on these 100 observations, is a Poisson distribution with a mean of 1.2 an appropriate model? Perform a goodness of fit procedure with $\alpha = 0.05$ ii) Calculate the P-value for this test	08
b	Ten samples are taken from a plating bath used in an electronics manufacturing process and the pH of the bath was determined. The sample pH values are 7.91, 7.85, 6.82, 8.01, 7.46, 6.95, 7.05, 7.35, 7.25 and 7.42. Manufacturing engineering believes pH has a median value of 7.0. i) Do the sample data indicate that this statement is correct? Use sign test with $\alpha = 0.05$ to investigate this hypothesis. Find the p-value for this test ii) Use the normal approximation for the sign test to test $H_o: \tilde{\mu} = 7.0 \; vs \; H_1: \tilde{\mu} \neq 7.0$	08
	OR	
9 a	An article is knee surgery considered arthoscopic meniscal repair with an absorbable screw. Results showed that for tears greater than $25mm$, 14 of 18 repairs were successful, but for shorter tears, 22 of 30 reapairs were successful. Is there evidence that the success rate is greater for longer tears? Use $\alpha = 0.05$. what is the p-value?	08

	b	Grades in a statistics course and an operations research course taken simultaneously were as follows for a group of students $Operation\ research\ grade$ $Statistics\ Grade$ | A | B | C | $Other$ A | 25 | 6 | 17 | 13 B | 17 | 16 | 15 | 6 C | 18 | 4 | 18 | 10 $Other$ | 10 | 8 | 11 | 20 Are the grades in statistics and operations research related? Use $\alpha = 0.01$ in reaching your conclusion.	08
10	a	In semiconductor manufacturing, wet chemical etching is often used to remove silicon from the backs of wafers prior to metallization. The etch rate is an important characteristic in this process and known to follow a normal distribution. Two different etching solutions have been compared using two random samples of 10 wafers for each solution. The observed etch rates are as follows $Solution\ I$ | 9.9 | 9.4 | 9.3 | 9.6 | 10.2 | 10.6 | 10.3 | 10.0 | 10.3 | 10.1 $Solution\ II$ | 10.2 | 10.6 | 10.7 | 10.4 | 10.5 | 10.0 | 10.2 | 10.7 | 10.4 | 10.3 i) Do the data support the claim that the mean etch rate is the same for both solutions? In reaching your conclusions, use $\alpha = 0.05$ and assume that both population variances are equal. ii) Find 95% confidence interval on the difference in mean etch rates.	08
	b	Derive normal equations for fitting regression equation $y = \beta_0 + \beta_1 x$ using method foe last squares.	08
		OR	
11	a	An article in the Journal of Environmental Engineering took a sample from the Holston River below king port, Tennessee during 1997. The biochemical oxygen demand (BOD) test is conducted over a period of time in days. The resulting data follow: $Time\ (days)$: | 1 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 BOD: | 0.6 | 0.7 | 1.5 | 1.9 | 2.1 | 2.6 | 2.9 | 3.7 | 3.5 | 3.7 | 3.8 i) Assuming that a simple linear regression model is appropriate, fit the regression model relating to BOD to the time. What is the estimate of σ^2? ii) What is the estimate of expected BOD level when the time is 15 days?	08
	b	Describe matrix approach to fit a multiple linear regression with usual notations.	08

12IM6C3

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

APPLIED STATISTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4 Answer FIVE full questions from Part B.
5 Use of statistical tables permitted.

PART A

1	1.1	Probability that at least 4 tosses of a fair coin are required for 3 heads is _____.	01
	1.2	There are 4 types of contaminants A, B, C and D found in a bottle of chemical. A is twice likely to be found as B, C and D are equally to be found, and each are twice as likely as A. Probability of C being found is _____.	02
	1.3	X is lognormal with parameters $\mu = 4$ and $\sigma^2 = 1.5$. Y is lognormal with parameters $\mu = 8$ and $\sigma^2 = 1.5$. $E(X)$ and $E(Y)$ are related as _____.	02
	1.4	Estimation techniques using sample data as well as information about parameters available beforehand are known as ___estimation.	01
	1.5	The likelihood function for a sample of 10 values drawn from a Poisson random variable with mean rate of 8 is _____.	01
	1.6	The test statistics used in a test for proportion of defectives based on 'd' defectives observed in parts is _____.	02
	1.7	The rejection region in a test for variance of a Normal population, $H_o: \sigma = \sigma_o$ vs $H_1: \sigma > \sigma_o$ is given by_____.	01
	1.8	What is the difference between contingency table test and goodness of fit tests?	02
	1.9	What is the model to be used to study the effect of interactions in a study with 2 independent variables?	01
	1.10	Which hypothesis is tested by the Wilcoxon Rank Sum test?	01
	1.11	What are the hypothesis statements in a test of significance of regression with 3 independent variables?	02
	1.12	How is multi collinearity among regressor variables estimated?	02
	1.13	The least square normal relations in matrix form is given as: _____ (mention content of each matrix)	02

PART B

2	a	Derive the moment generating function $M_X(t)$ of a random variable X representing the no. of successes in a set of n trials, with probability p of a trial being a success. Using $M_X(t)$, derive the expectation and variance of X.	08

	b	A random variable X has its probability density function as $f(x) = (1/2)x^2e^{-x},\ x > 0$. Find its moment-generating function $MX(t)$ and hence find its expression.	08
		OR	
3	a	X is equally likely between the interval 1 and 2. How is $Y = e^{-x}$ distributed? Compute the probability of Y being at most 0.25. find the expectation of Y.	08
	b	Suppose a random variable Y has the probability density function given by $f(y) = 6y(1-y)$ for $0 \leq y \leq 1$. Find $F(y)$, its mean and median. Find $P(0.5 \leq y \leq 0.8)$	08
4	a	Suppose $Mx(t) = 9/(3-t)^2$. Derive an expression for $E(x^n)$. What is the $var(x)$.	08
	b	Proportion of down-time of a machine, P is known to be Beta distributed with probability density $f(p) = 2(1-p)$. Find the expectation and variance of the cost of down-time W (due to lost production, maintenance and re-set up costs), given by $W = 10 + 20P$.	08
		OR	
5	a	The life time of a projector lamps follows a log-normal distribution with a co-efficient of variation (ratio of std dev to mean) of 2. The expected life time is 5000 hours of operation. i) Determine the parameters of the distribution and the probability that a lamp will last 6000 hours. ii) How much of is the lifetime 95% of the lamps exceed?	08
	b	The time between severe earthquakes at a given region follows a log-normal distribution with a coefficient of variation (ratio of std dev to mean) of 40%. The expected time between severe earthquakes is 80 years. i) Determine the parameters of this log-normally distributed interval time T. ii) Determine the probability that a severe earthquake will occur within 20 years from previous one. iii) Suppose the last severe earthquake in the region took place 100 years ago. What is the probability that a severe earthquake will occur the next year.	08
6	a	Determine the value of c that makes the function $f(x,y) = ce^{(-2x-3y)}$ a joint probability density function over the range $0 < x < y$. Determine the following: i) Marginal probability distribution of X. ii) Conditional probability distribution of Y given $X = 1$. iii) $E(Y\|X = 1)$ iv) Are X and Y independent? Why or why not?	08
	b	Two methods of measuring surface smoothness are used to evaluate a paper product. The measurements X and Y are recorded as deviations from the normal surface smoothness in coded units. The joint probability distribution of the two measurements is a uniform distribution over the region $0 < x < 4, 0 < y$ and $x - 1 < y < x + 1$.	

		Determine the following: i) $P(X < 0.5, Y < 0.5)$ ii) $P(X < 0.5)$ iii) Marginal probability distribution of X iv) Conditional probability distribution of Y given $X = 1$. v) $E(Y/X = 1)$	08
		OR	
7	a	Three electronic ovens that were dropped during shipment are inspected and classified as containing either a major, a minor, or no defect. In the past, 60% of dropped ovens had a major defect, 30% had a minor defect, and 10% had no defect. Assume that the defects on the three ovens occur independently. i) What is the probability distribution of the count of ovens in each category ii) Derive the joint probability mass function of the number of ovens with different types of defects. iii) Compute the expected number of ovens with a major defect.	08
	b	i) Let X be a geometric random variable with parameter p. find the maximum likelihood estimator of p, based on a random sample of size n. ii) $\overline{Y_1}$ and S_1^2 are the sample mean and sample variance from a population with mean μ_1 and variance σ_1^2. Similarly $\overline{Y_2}$ and S_2^2 are the sample mean and sample variance from a second independent population with mean μ_2 and variance σ_2^2. The sample sizes are n_1 and n_2, respectively. Show that $\overline{Y_1}$- $\overline{Y_2}$ is an unbiased estimators of $\mu_1 - \mu_2$. Find the standard error of $\overline{Y_1} - \overline{Y_2}$.	08
8	a	A company operates four machines three shifts each day. From production records, the following data on the number of breakdowns are collected: $Shift$ | $Machines$: A | B | C | D 1 | 41 | 20 | 12 | 16 2 | 31 | 11 | 9 | 14 3 | 15 | 17 | 16 | 10 Test the hypothesis at 5% significance that breakdowns are independent of the shift. How will you find the P-value for this test?	08
	b	The titanium intent in an aircraft-grade alloy is an important determinant of strength. A sample of 20 test coupons reveals the following titanium content(in percent): 8.32,8.05,8.93,8.65,8.25,8.46,8.52,8.35,8.36,8.41,8.42,8.30,8.71,8.75,8.60,8.83, 8.50,8.38, 8.29,8.46 i) The median titanium content should be 8.5%. Use the sign test with significance of 0.05 to investigate the hypothesis. ii) Suppose that the distribution of titanium content is symmetric and continuous. Use Wilcoxon signed-rank test to test if median is 8.5%.	08
		OR	

9 a Grades in a statistics course and an operation research course taken simultaneously were as follow for a group of students

Grade in statistics	*Operation research grade*			
	A	*B*	*C*	*other*
A	25	6	17	13
B	17	16	15	6
C	18	4	18	10
Other	10	8	11	20

Are the grades in statistics and operation research related? Use 1% significance in reaching your conclusion. 08

b Ten samples were taken from plating bath used in an electronics manufacturing process, and the bath pH was determined. The sample pH values are $7.91, 7.85, 6.82, 8.01, 7.46, 6.95, 7.05, 7.35, 7.25, 7.42$. Manufacturing engineering believes that pH has a median value of 7.0. Do the sample data indicate that this statement is correct? Use the sign test with 5% significance to investigate this hypothesis. How will you find the p-value for this test? Use the normal approximation for the sign test to see if the median is different from a pH value of 7.0. 08

10 a Two machines are used for filing bottles with a net volume of $16.0\,ml$ of a drug. The fill volume can be assumed normal, with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025ml$. A quality engineering staff suspects that both machines fill to the same mean net volume of $16.0\,ml$. a random sample of 10 bottles is taken from the output of each machine.

Machine 1		*Machine* 2	
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

i) Do you think the engineer is correct? Use $\alpha = 0.05$
ii) What is the p-value for this test?
iii) Find 95% confidence interval on the difference in means. Provide a practical interpretation of this interval. 08

b Enumerate the various techniques employed in variable selection in multiple linear regressions. How are the techniques to choose the best set of variables? 08

OR

11 a A procurement specialist has purchased 25 resistors from vendor1 and 35 resistors from vendor 2. Each resistror's resistance is measured with the following results:
Vendor1: $96.8, 100.0, 100.3, 98.5, 98.3, 98.2, 99.6, 99.4, 99.9, 101.1, 103.7, 97.7, 98.6$
Vendor2:$106.8, 106.8, 104.7, 104.7, 108.0, 102.2, 103.2, 103.7, 106.8, 105.1, 104.0, 106.2, 102.6, 100.3, 104.0, 107.0, 104.3, 105.8$
Perform an appropriate statistical hypothesis-testing procedure to determine whether the procurement specialist can claim that the variance of resistance of product from vendor 1 is significantly different from variance of resistance of product from vendor 2. 08

b How do you investigate whether an independent variable affects the output or not? Explain with the help of an example, using multiple regression techniques and indicator variables. 08

12IM6C3

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

APPLIED STATISTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	When shape and scale parameters of a Gamma distribution are 3 and 4.5 respectively, its expectation is _____.	01
	1.2	When the parameters of a Beta distribution are equal, its expectation is _____.	01
	1.3	The random variable representing the number of tosses of a coin required to get 3 Heads, is distributed as _____ distribution.	01
	1.4	For a random variable x, its variance can be expressed in terms of its moment generating function $M_x(t)$ as _____.	01
	1.5	X is lognormal with parameters $\mu = 4$ and $\sigma^2 = 1.5$ the mean and variance of X are _____ and _____ respectively.	02
	1.6	In a lot of 850 parts, 50 are defective. 10 parts are chosen without replacement for inspection. The expectation and variance of the number of defective parts out of 10 are _____ and _____ respectively.	02
	1.7	The test statistic used in the signed rank test is computed as _____. (mention what the terms mean)	01
	1.8	The likelihood function for a sample of 10 values drawn from a Poisson distributed population with a mean of 8 is _____. (The sample is $x_1, x_2, \ldots\ldots\ldots x_{10}$)	01
	1.9	An inspection process involves picking 20 parts and classifying them as Good/Fair/Poor. The number of ways of sequencing 9 Good, 8 Fair and 3 Poor parts for further analysis is _____.	01
	1.10	The covariance between two variables X and Y can take on values in the range _____.	01
	1.11	When can a paired t-test be used when comparing the means of two populations?	01
	1.12	The test statistic used to compare variances of two populations is _____.	01
	1.13	What do the diagonal entries of the covariance matrix contain?	01
	1.14	The error degree of freedom in a multiple linear regression with 50 observed values y_i and 7 regressor variables is _____.	01
	1.15	Type II error is committed when the conclusion is _____ the null hypothesis when the null hypothesis is _____.	01

	1.16	What are the hypothesis statements in a test of significance of regression, involving two independent variables?	01
	1.17	What is the pooled variance in a two-sample test drawn with sample sizes $n_1$and n_2, and when the population variances are unknown but equal?	02

PART-B

2	a	How is the moment generating function $M_x(t)$ defined for a discrete distribution and how is it used to compute the mean and variance?	06
	b	Given a moment generating function $M_x(t) = \frac{1}{8}e^{-5t} + \frac{1}{4}e^{t} + \frac{3}{8}e^{7t} + k$, compute the value of k and write the mass function. Derive $E[X^n]$, and hence find the mean and variance.	10
		OR	
3	a	The overtime T in hours, in a production shop floor varies continuously between 0 to 4 hours, with the pdf $(t) = \frac{t}{8}$. The total wages w, including the base pay is $200\,t + 400$. Find the pdf of W, $f(w)$.	08
	b	In a game, a circular hoop is thrown towards a few poles at a distance. The hoops land over a pole with a probability p. Players keep throwing hoops one by one till a hoop lands over a pole. The cost of the game $= X^2$, where X = number of throws of the hoop. Find the probability distribution of C, $f_c(c)$.	08
4		A state run lottery requires 3 numbers to be chosen from 20 numbers without replacement. A jackpot is won if all 3 chosen numbers match the state's sample.	
		a) What is the probability of a jackpot win?	03
		b) If a player plays this lottery every week, what is the expected number of weeks to play to hit a jackpot win?	03
		c) What is the probability that the player achieves at least one matching within 3 weeks of play?	10
		OR	
5		Lifetime of an electronic gadget used in a communication equipment is log normally distributed with parameters $\mu = 10$ hrs and $\sigma = 1.5$ hrs (not the mean and standard deviation)	
		a) What is the probability that the lifetime exceeds 10.000 hours?	04
		b) What is the lifetime exceeded by 99% of the gadgets?	05
		c) What is the probability that exactly 5 gadgets have to be tested for an inspector to encounter a gadget having a lifetime in excess of 20,000 hours?	07

6 A company uses two different inspection devices to test 4 randomly selected parts. The first device detects defects 99.3% of the time and the second, 99.7% of the time. The devices operate independently of each other. Let X and Y denote the number of defective parts identified by device 1 and 2 respectively. Determine:

a) Range of values of X and Y. Are X and Y independent? Why or why not? 02

b) The joint probability mass function $f_{X,Y}(x,y)$, as an expression 02

c) Marginal probability mass functions $f(x)$ and $f(y)$. 04

d) Expectation and variance of Y, given that device 1 has identified 2 parts as defective. 04

e) What is the probability that the 1st device identifies 2 defectives and 2nd identifies 3 defectives? 04

OR

7 Suppose $X_1, X_2, \ldots X_n$ is a random sample of size n drawn from a normal distribution with unknown mean and variance μ and σ^2 respectively. $f(x) = \frac{1}{\sigma\sqrt{2\pi}} l^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $-\infty < x < \infty$. Derive and compare estimators for μ and σ^2 using

a) Method of moments, and

b) Method of maximum likelihood. 16

8 a Eleven identical units of an electronic hardware were tested and number of hours of continuous operation before overheating and shutting down was recorded as: $1.5, 2.2, 0.9, 1.3, 2.0, 1.6, 1.8, 1.5, 2.0, 1.2, 1.7$ Test at 5% level whether the mean length of continuous operation is 1.8 hours or not. Use Wilcoxon signed rank test. 08

b Strength measurements on 9 specific steel plates used in construction were made using two methods for comparison, as shown below:

Plate	1	2	3	4	5	6	7	8	9
*Method*1	1.19	1.15	1.32	1.34	1.2	1.4	1.37	1.54	1.56
Method 2	1.06	0.99	1.06	1.06	1.07	1.18	1.04	1.09	1.05

i) Construct a 95% confidence interval on the difference in strength from the two methods.

ii) Test if there is any difference between the two methods, at 5% significance. Is the result consistent with the findings of part (i). Explain why. 08

OR

9 a Failure data for electronic devices has the following observations categorized according to type of failure and mounting positions for 134 devices:

Failure Type		*A*	*B*	*C*	*D*
Mounting Portion	1	22	46	18	9
	2	4	17	6	12

Is failure type independent of mounting position? Test at 1% significance. 06

b	In order to evaluate on time performance of cab companies A and B, the number of minutes late is recorded for the last 6 journeys of each company as below in Table 9(b). Negative values indicate early arrival. Is there evidence that either company has better on-time performance than the other? Test using Wilcoxon rank-sum test at 1% significance.	06
c	In a random sample of 85 bearings, 10 have poor surface finish. Construct a 95% 2-sided confidence interval for the population proportion of bearings with poor surface finish.	04
10	An analyst has collected information about an output y for different settings of x, and the summary quantities are: $n = 20$, $\sum x_i = 23.92$, $\sum y_i = 1843.21$, $\sum x_i^2 = 29.29$, $\sum y_i^2 = 170044.21$, $\sum x_i y_i = 2214.66$ a) Estimate the slope and intercept assuming a linear relation b) Test for significance of regression, using $ANOVA$, at 1% significance c) The analyst suspects that the true linear relation involves a slope of $\beta_1 = 12$. Test the claim at 1% significance.	04 06 06
	OR	
11	A construction firm studied soil conditions depth in feet x_1, and moisture content in %, x_2, affecting the shear strength y of the soil. Summary quantities are computed from 10 observations as follows: $\sum x_{i1} = 223, \sum x_{i2} = 553, \sum x_{i1}^2 = 5200.9, \sum x_{i2}^2 = 31729, \sum x_{i1}x_{i2} = 12352, \sum y_i^2 = 371595.6, \sum x_{i1}y_i = 43550.8 \sum x_{i2}y_i = 104736.8$ Few entries of the matrix $(X'X)^{-1}$ are: $\begin{bmatrix} 4.76 & -.09 & -.05 \\ & .004 & 7.7 \times 10^{-5} \\ & & .009 \end{bmatrix}$ a) Write the normal equations for least square regression b) Estimate coefficients of regression $\beta_0, \beta_1, \beta_2$ c) Estimate the soil shear strength at 18 feet with 43% moisture.	08 06 02

12IM6C3

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

APPLIED STATISTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	Determine the range (or possible values) of the random variable in the number of options selected while ordering for an automobile which is sold in the market with a base model and the options to add any number of 15 optional features.	01
	1.2	Suppose $f(x) = x/8$ for $3 < x < 5$, determine the mean and variance for X.	02
	1.3	If X and Y are independent, normal random variables with $E(X) = 0, V(Y) = 4, E(Y) = 10, V(Y) = 9$. Determine $E(2X + 3Y)$.	01
	1.4	The number of times a transistor in a computer memory changes state in one operation is a _____ random variable.	01
	1.5	If X denote the number of trials until the first success. Then X is a _____ random variable with parameters $0 < p < 1$ and $f(x) = (1-p)^{x-1} * p$ for $x = 1,2 \ldots$	01
	1.6	If X is a negative binomial random variable with parameters p and r, then the mean of the random variable $\mu = E(X) =$ _____.	01
	1.7	The net weight of a packaged chemical herbicide is uniform for $49.75 < X < 50.25$ pounds. The mean of the weight of the packages is _____.	01
	1.8	If X is a gamma random variable with parameters λ and γ then the mean of the random variable is given as $\mu = E(X) =$ _____.	01
	1.9	If the weibull distribution has parameters δ(scale parameters) $\delta > 0$ and shape parameters $\beta(\beta > 0)$ then the cumulative distribution is given by the expression $F(x) =$ _____.	01
	1.10	The random variables $X_1, X_2 \ldots X_K$ that denote the number of trials that result in class1,class2…class$k$ respectively have a _____ distribution with a defined joint probability mass function.	01
	1.11	A point estimate of some population parameter θ is a single numerical value θ of a statistic θ. The statistic θ is called the _____.	01
	1.12	If we consider all unbiased estimators of θ, then the one with smallest variance is called the _____ ($MVUE$ for short).	01
	1.13	The function $L(\theta) = f(x_1:\theta).f(x_2:\theta)\ldots..f(x_n:\theta)$ is known as the _____ function of the sample.	01
	1.14	The _____ value is the smallest level of significance that would lead to reflection of the null hypothesis H_0 with the given data.	01

	1.15	The test statistic for the Goodness of fit test on distribution is given as $X_O^2 =$ _____, where O_i is the observed frequency and E_i is the expected frequency in the ith class.	01
	1.16	If data is collected in pairs and we need to compare the differences between the pair of valves, then a special case of t test called the _____ is applied.	01
	1.17	Regression models are constructed based on data collected from the field or the process, this data driven modeling approach is known as _____ model.	01
	1.18	The residuals $e = y_i - \hat{y}_i$ are used to obtain an estimate of σ^2. The sum of squares of the residuals is often called the _____, given by the expression $\delta\delta_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$.	01
	1.19	The two component source of variation, when the analysis of variance procedure is applied on the regression model are the _____ sum of squares and the Error sum of squares.	01

PART-B

2	a	In an automated filling operation, the probability of an in correct fill when the process is operated at a low speed is 0.001. when the process is operated at high speed the probability of an incorrect fill is 0.01. assume that 30% of the containers are filled when the process is operated at a high speed and the remainder are filled when the process is operated at a low speed. i) What is the probability of an incorrectly filled container? ii) If an incorrectly filled container is found, what is the probability that it was filled during the high speed operation?	06
	b	An assembly consists of three mechanical components. Suppose that the probabilities that the first, second and third c components meet specifications are 0.95 0.98 and 0.99. Assume that the components are independent. Determine the probability mass function of the number of components in the assembly that meet specifications.	06
	c	Define a Moment generating function. How do you determine the moments using the moment generating function? Give an example.	04
		OR	
3	a	A manufacturer of a consumer electronics product expects 2% of units to fail during the warranty period. A sample of 500 independent units is tracked for warranty performance i) What is the probability that none fails during the warranty period? ii) What is the expected number of failures during the warranty period? iii) What is the probability that more than two units fail during the warranty period?	06
	b	The lifetime of a semiconductor laser has a log-normal distribution and it is known that the mean and standard deviation of lifetime are 10,000 hours and 20,000 hoursrespectively. i) Calculate the parameters of the log normal distribution ii) Determine the probability that a life time exceeds 10,000 hours. iii) Determine the lifetime that is exceeded by 90% of lasers.	06

	c	Assume that the life of a roller bearing follows a Weibull distribution with parameters $\beta = 2$ and $\delta = 10{,}000$hours. i) Determine the probability that a bearing lasts at least 8000 hours. ii) Determine the mean time until failure of a bearing.	04
4	a	It is known that two defective copies of a commercial software program were erroneously sent to a shipping lot that has how a total of 75 copies of the program. A sample of copies will be selected from a lot without replacement. i) If three copies of the software are inspected determine the probability that exactly one of the defective copies will be found ii) If three copies of the software are inspected, determine the probability that both defective copies will be found.	08
	b	Let the random variables X_1 and X_2 denote the length and width respectively of a manufactured part. Assume that X_1 is normal with $E(X_1) = 2$ centimeters and standard deviation 0.1 centimeters and that X_2 is normal with $E(X_2) = 5$ centimeters and standard deviation 0.2 centimeters. Also assume that X_1 and X_2 are independent. Determine the probability that the perimeter exceeds 14.5 centimeters.	08
		OR	
5	a	Suppose that the log-ins to a computer network follow a poisson process with an average of 3 counts per minute i) What is the mean time between counts? ii) What is the standard deviation of the time between counts? iii) Determine the value of x such that probability that atleast one count occurs before time x minutes is 0.95.	08
	b	Assume that life of a packaged magnetic disk exposed to corrosive gases has a weibull distribution with $\beta = 0.5$ and the mean life of 600 hours. i) Determine the probability that a packaged disk last atleast 500 hours. ii) Determine the probability that a packaged disk fails before 400 hours.	06
	c	What are the uses of moment generating function?	02
6	a	Four electronic ovens that were dropped during shipment are inspected and classified as contacting either a major, minor or no defect. In the past 60% of dropped ovens had a major defect, 30% had a minor defect and 10% had no defect. Assume that the defects on the four ovens occur independently. i) Is the probability distribution of the count of ovens in each category multinomial? Why or why not? ii) What is the probability that of the four dropped ovens two have a major defect and two have a minor defect? iii) What is the probability that no oven has defect?	08

	b	Suppose we have a random sample of size $2n$ from a population denoted by X_1 and $E(X) = \mu$ and $V(X) = \sigma^2$. Let $\bar{X}_1 = \frac{1}{2n}\sum_{i=1}^{2n} X_i$ and $\bar{X}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i$ be two estimates of μ. Which is the better estimator of μ. Explain your choice.	06
	c	Define the likelihood function.	02
		OR	
7	a	The weight of adobe bricks used for construction are normally distributed with a mean of 3 pounds and a standard deviation of 0.25 pound. Assume that the weights of the bricks are independent and that a random sample of 20 bricks is selected. i) What is the probability that all the bricks in the sample exceeds 2.75 pounds? ii) What is the probability that the heaviest brick in the sample exceeds 3.75 pounds?	08
	b	Suppose that $\widehat{\theta_1}$and $\widehat{\theta_2}$ are unbiased estimators of the parameter θ. We know that $V(\widehat{\theta_1}) = 10$ and $V(\widehat{\theta_2}) = 4$. Which estimator is best and in what sense it is best?	04
	c	Explain the method of moments as an estimating procedure?	04
8	a	A manufacturer of electronic calculator takes a random sample of 1200 calculators and finds that there are eight defective units. i) Construct a 95% confidence interval on the population proportion. ii) Is there a evidence to support a claim that the fraction of defective units produced is 1% or less?	06
	b	Two companies manufacture a rubber material intended for use in automotive application. The part will be subject to abrasive wear in the field application so we decide to compare the material produced by each company in a test. Twenty five samples of material from each company are tested in an abrasion test and the amount of wear after 1000 cycles is observed. For company 1 the sample mean and standard deviation of wear are $\overline{X_1} = 20$ milligram/1000cycles and $s_1 = 2$ milligram/1000 cycles, while for company 2 we obtain $\overline{X_2} = 15$ multigrams/1000 cycles and $s_2 = 8$ milligram /1000 cycles i) Do the data support the claim that the two companies produce material with different mean wear? Use $X = 0.05$ and assume each population is normally distributed but the variances are not equal? ii) What is the P-value of the test?	10
		OR	
9	a	The manager of a fleet of automobiles is testing two brands of radial tires and assigns one size of each brand at random to the two rear wheels of eight cars and runs the cars until the tyres wear out.	

Car	$Brand\ 1$	$Brand\ 2$
1	36,925	34,318
2	45,300	42,280
3	36,240	35,500
4	32,100	31,950
5	37,210	38,015
6	48,360	47,800
7	38,200	37,810
8	33,500	33,215

The data (in kilometers) follow. Find a 99% confidence interval on the difference in mean life. Which brand would you prefer based on this calculation. 08

b Oxide layers on semiconductor wafers are etched in a mixture of gases to achieve the proper thickness. The variability in the thickness of these oxide layers is a critical characteristic of the wafer and low variability is desirable for subsequent processing steps. Two different mixture of gases are being studied to determine whether one is superior in reducing the variability of oxide thickness. Twenty wafers are etched in each gas. The sample standard deviation of oxide thickness are $s_1 = 1.96$ angstoms and $s_2 = 2.13$ angstroms respectively. Is there any evidence to indicate that either gas is preferable? Use $\alpha = 0.05$. 08

10 The grams of solid removed from a material (y) is thought to be related to the drying time. Ten observations obtained from an experimental study follow.

y	4.3	1.5	1.8	4.9	4.2	4.8	5.8	6.2	7.0	7.9
x	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0

a) Construct a scatter diagram for these data
b) Fit a simple linear regression model
c) Test the significance of regression
d) Based on the data, what is your estimate of the mean grams of solid removed at 4.25 hours? Find a 95% confidence interval on the mean
e) Analyze the residuals and comment on the model adequacy. 16

OR

11 Simple regression model was used to express drain currently(in milli amperes) as a function of ground-to-source voltage x in volts in the data obtained are as follows:

y	x	y	x
0.734	1.1	1.50	1.6
0.886	1.2	1.66	1.7
1.04	1.3	1.81	1.8
1.19	1.4	1.97	1.9
1.35	1.5	2.12	2.0

a) Draw a scatter diagram of these data. Does a straight line relationship seem possible?
b) Fit a simple linear regression model to these data?
c) Test for significance of regression using $\alpha = 0.05$. What is the P-value for this test?
d) Find a 95% confidence interval estimate on slope.
e) Test the hypothesis $H_o\beta_o = 0$ versus $H_1\beta_o \neq 0$ using $\alpha = 0.05$. What conclusions can you draw? 16

Enterprise Resource Planning and E-Commerce

VI Semester SEE Question Papers

Course codes: 12IM6D5

- Apr/May 2018 (12IM6D5)
- Apr/May 2017 (12IM6D5)
- Fast Track Jul 2016 (12IM6D5)
- Apr/May 2016 (12IM6D5)

12IM6D5

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

ENTERPRISE RESOURCE PLANNING AND E-COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Differentiate between traditional commerce and electronics commerce.	02
	1.2	What are the characteristics of supply chain management in an e-commerce environment?	02
	1.3	What are the types of security features used in client server types of network?	02
	1.4	Why firewalls are used?	02
	1.5	Expand the acronyms EFT and EDI.	02
	1.6	Differentiate between $MRP-I$ and $MRP-II$.	02
	1.7	List the categories of e-commerce.	02
	1.8	Explain the application of e-commerce in the home banking.	02
	1.9	Define e-commerce. What are the benefits of using e-commerce?	02
	1.10	Describe the process of data mining.	02

PART B

2	a	Define ERP and explain the importance of ERP.	08
	b	Explain the methodology used in ERP implementation.	08
		OR	
3	a	Do you justify ERP investments in startup companies? Give examples.	08
	b	"Supply Chain is vastly benefited by enabling ERP". Explain by providing illustrations.	08
4	a	Discuss online analytical processing with suitable examples. Sketch a neat block diagram.	08
	b	Identify and analyze various internet business models with examples.	08
		OR	

5	a	How relevant is Business Process Reengineering in today's context? Explain different phases with their advantages and examples.	08
	b	Identify and explain any four modules of *ERP* system with suitable examples.	08
6	a	What are the problems faced during *ERP* implementation? How to overcome them? Give examples.	08
	b	Explain the life cycle phases of the *ERP* implementation with illustrations.	08
		OR	
7	a	Evaluate the critical factors involved in successful implementation of *ERP* and highlight their impact.	08
	b	Assess the post *ERP* implementation activities with suitable examples. How important are these activities?	08
8	a	Explain the relevance of e-commerce in today's markets. Explain with examples.	08
	b	Differentiate between digital market and traditional market by considering competitive factors.	08
		OR	
9	a	List and explain the components of E-commerce. Illustrate with an example.	08
	b	Explain how different departments participate in an e-commerce project. Is it right from the systems point of view?	08
10	a	Identify the nine essential e-commerce processes required for the successful operation or management of e-commerce activities.	08
	b	Evaluate some of the trends of e-commerce applications with examples.	08
		OR	
11	a	Analyze the benefits and challenges of e-commerce.	08
	b	How do you rate the e-commerce activities from the Buyer's experience? Give suitable examples.	08

12IM6D5

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

ENTERPRISE RESOURCE PLANNING AND E-COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Differentiate between the business function and business process.	02
	1.2	What is subject oriented, integrated, non-volatile & time-variant data warehouse?	02
	1.3	Give one example each for a proprietary & a open source ERP software.	02
	1.4	One of the most critical step in the ERP implementation is the creation of an _____.	01
	1.5	The minimal information system consists of three elements _____, _____ and _____	02
	1.6	List the objective of ERP implementation.	02
	1.7	Give one example each for $B2C$ & $B2E$ e-business transaction models.	02
	1.8	Draw the single block diagram showing the relation between $B2C$ & $B2E$ models.	02
	1.9	Differentiate between $OLAP$ and $OLTP$.	02
	1.10	List the sales related business transaction.	02
	1.11	_____ Module facilitates the process of maintaining the appropriate level of stock in a warehouse.	01

PART B

2	a	"Thinking in terms of business process helps mangers to look at their organizations from a customer's perspective:. Explain with a real life example.	08
	b	List the common myth about the ERP system and demystify them.	08
		OR	
3	a	Summarize the evolution of ERP from $1960s$ to $1990s$.	08
	b	Highlight different reasons for the growth of ERP market.	08

4	a	Explain the integration of ERP, supply chain and customers relationship applications with the help of neat sketch.	10
	b	Why is $OLAP$ is called fast analysis & shared multi-dimensional information.	06
		OR	
5	a	How do you address different security issues involved in ERP implementation?	08
	b	Bring out the difference between data mining and data warehouse system.	08
6	a	Build a business model of a ERP system for an education institution like $RVCE$.	10
	b	List and explain the various skills required by the people who are involved in the ERP implementation.	06
		OR	
7	a	Develop a framework to implement ERP system in a health care organization.	08
	b	Choosing the right ERP vendors goes beyond evaluating the software functionality. Explain.	08
8	a	Explain most widely used client server applications.	08
	b	Compare different e-business transaction models with suitable examples.	08
		OR	
9	a	With the help of a block diagram explain internet administrative structure.	08
	b	What are the important concepts introduced in the IT Act, 2000.	08
10	a	Compare traditional marketing over e-marketing.	08
	b	Summarize the general security issues that e-business must consider.	08
		OR	
11	a	Discuss the different components of m-commerce.	08
	b	Recommend a marketing strategy for m-commerce for a restaurant at your city.	08

12IM6D5

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

ENTERPRISE RESOURCE PLANNING AND E - COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	_____ is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical, contemporary measures of performance such as cost, quality, service and speed.	01
	1.2	Define $OLAP$.	02
	1.3	Differentiate between business function and process.	02
	1.4	The extraction of hidden predictive information from large database is called as _____.	01
	1.5	What is the subject oriented, integrated, non-volatile and time variant data warehouse?	02
	1.6	Define $OLTP$.	02
	1.7	Classify different E-business transaction models.	02
	1.8	What are the important concepts introduced in the IT act, 2000.	02
	1.9	What is digital signature?	01
	1.10	Establish relation between $B2B$ and $B2C$ models.	02
	1.11	What dynamic IP addressing.	02
	1.12	Define e-supply chain management.	01

PART-B

2	a	Explain with examples why it is important to have interdepartmental communication for the smooth and efficient functioning of an organization.	08
	b	What are the problems in ERP implementation according to survey finding reported in the literature?	08
		OR	
3	a	What are the common myths about ERP implementation?	08
	b	"Thinking in terms of business processes helps managers to look at their organizations from a customer perspective". Explain with example.	08

4	a	Discuss different tools and techniques used in data mining.	08
	b	Define BPR. List different phases involved in BPR implementation. Explain any two phases in detail.	08
		OR	
5	a	Develop a flow for sales and distribution module for a garment company.	08
	b	Explain the integration of ERP, supply chain and customer relationship applications with the help of neat sketch.	08
6	a	Develop a questionnaire containing minimum ten questions to collect the requirements of customers with respect to ERP software.	08
	b	Evaluate the pros and cons of proprietary and open source ERP implementation.	08
		OR	
7	a	Create a test plan to evaluate the effectiveness of ERP implementation. Give two examples showcasing your test cases.	08
	b	Select right implementation strategy for organization having different business processes. Justify your answer.	08
8	a	Plan a survey to evaluate the popularity of e-commerce over traditional commerce.	08
	b	Compare different e-business transaction models with suitable examples.	08
		OR	
9	a	Prioritize the key challenges that exist in the e-commerce industries.	08
	b	What are the processes involved in $B2C$ transaction model?	08
10	a	Explain the client server applications of internet.	08
	b	Recommend a marketing strategy for m-commerce business model.	08
		OR	
11	a	Prepare a list of factors governing E-Security. How these factors can be addressed.	08
	b	Evaluate the effectiveness of internet standards and specifications in Indian context.	08

12IM6D5

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

ENTERPRISE RESOURCE PLANNING AND E-COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	_____ is a collection of activities that takes one or more kinds of input and creates a output that is of value to the customer.	01
	1.2	The first phase of ERP implementation is _____	01
	1.3	What is subject oriented, integrated, non-volatile and time variant data warehouse?	02
	1.4	_____ are responsible for fixing any problems in software that the implementation team encounters.	01
	1.5	GIS stands for _____.	01
	1.6	What are the characteristics of the information?	02
	1.7	List any four concepts introduced in the IT Act, 2000 of government of India.	02
	1.8	Give one example for $B2C$ transaction model.	01
	1.9	Expand FTP.	01
	1.10	Establish a relationship between $B2B$ and $B2C$ models using a block diagram.	02
	1.11	_____ is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical, contemporary measures of performance such as cost, quality, service and speed.	01
	1.12	_____ is the process of identifying valid, Novel, potentially useful, and ultimately comprehensive knowledge from database that is used to make crucial business decisions.	01
	1.13	_____ module streamline the management of human resources and human capital.	01
	1.14	Define e-supply chain management.	02
	1.15	What is M-commerce?	01

PART-B

2	a	Explain with examples why it is important to have interdepartmental communication for smooth and efficient functioning of an organization.	08
	b	What is meant by Integrated Data Model? Explain how it is created.	08

OR

3	a	Many enterprises are replacing their legacy business systems with integrated versions. Elaborate on the advantages of installing ERP systems in the enterprises.	08
	b	Provide an overview on the evolution of ERP systems.	08
4	a	Define BPR. List different phases involved in BPR implementation. Explain phase one in detail.	08
	b	Bring out the differences between online analytical and online Transactional processing systems.	08
		OR	
5	a	Explain the functions, subsystems and features of the sales and distribution module in ERP system.	08
	b	What are the functions, subsystems and features of the HR module in ERP systems?	08
6	a	Explain why it is not a good idea to develop an ERP package in-house.	08
	b	Choosing the right ERP vendor goes beyond evaluating the software functionality. Explain.	08
		OR	
7		Discuss with the help of a schematic representation, the different phases of ERP implementation life cycle.	16
8	a	List the key strengths of using the internet for business in India.	08
	b	Compare different e-business transaction models with suitable examples.	08
		OR	
9	a	Explain through a flow chart the processes involved in $B2C$ e-commerce.	08
	b	Prioritize the key opportunities that exist in the e-commerce industries.	08
10	a	List and explain the client-server protocols of internet.	08
	b	Recommend a marketing strategy for M-commerce business model.	08
		OR	
11	a	Prepare a list of factors governing e-security. How these factors can be addressed?	08
	b	Evaluate the effectiveness of internet standards and specification in India context.	08

Management Information Systems

VI Semester SEE Question Papers

Course codes: 12IM6C5

- Apr/May 2018 (12IM6C5)

12IM6C5

USN | | | | | | | | | | |

R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

MANAGEMENT INFORMATION SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Identify the fundamental roles of *IS* in Business.	02
	1.2	What are the steps involved in developing Information Systems Solutions?	02
	1.3	What are the Information System Resources?	02
	1.4	List different developments in many types of storage media and devices.	02
	1.5	What are the five fundamental database structures?	02
	1.6	What are the major telecommunications trends?	02
	1.7	What are the categories of major financial management systems?	02
	1.8	What are the different levels of managerial decision making that must be supported by information technology in a successful organization?	02
	1.9	What is *OLAP*?	02
	1.10	What is "what if analysis"?	02

PART B

2	a	Explain why knowledge of information systems is important for business professionals. Identify and discuss with neat sketch five areas of information systems knowledge that they need.	08
	b	Discuss with neat sketch several major types of information systems from your experiences that support business operations and managerial decision making at business organizations in the real world.	08
		OR	
3	a	Continued developments in information technology and its business applications have resulted in the evolution of several major types of databases. As an *MIS* manager illustrate several major conceptual categories of databases that may be found in many organizations.	08
	b	As businesses continue to use some open technologies to improve communication with customers and partners, they can gain many competitive advantages along the way in product development, cost savings, marketing, distributions and leveraging their partnerships. Which type of internet technology do you suggest as MIS manager? Discuss with neat sketch how the business will derive the benefits.	08
4	a	Briefly discuss with a neat sketch, how the major functional areas of business are supported by in functional business information systems.	08

	b	Discuss how the major application components of enterprise resource planning demonstrate the cross functional approach of *ERP* systems and also show different flows that support *ERP* systems.	08
		OR	
5	a	Identify the essential processes of an e-commerce system and give examples of how it is implemented in e-commerce applications.	08
	b	What are e-Commerce market places? Identify and discuss the different types of e-commerce market places used by business today.	08
6	a	Enumerate the major differences in the information and decision support capabilities of management information systems and decision support systems.	08
	b	Describe how online analytical processing can meet key information needs of managers.	08
		OR	
7		Briefly discuss how the following emerging areas of decision support systems support decision making in business: a) Neural network b) Expert systems c) Fuzzy logic d) Genetic algorithm.	16
8	a	Betting on new *IT* innovations mean betting the future of the company. Leading edge firms are sometime said to be on the "bleeding edge". Hence, planning for competitive advantage is especially important in today's competitive business arena and complex information technology environment. So, strategic business/*IT* planning involves an evaluation of the potential benefits and risks a company faces when using *IT*-based strategies and technologies for competitive advantage. In such situation, how do you proceed for planning *IT* strategies? Discuss.	08
	b	"Business model" was one of the great buzzwords of the internet boom, routinely invoked, as the writer Michael Lewis puts it, "to invoke all manner of half-baked plans". A good business model however remains essential to every successful organization, whether it's a new venture or an established player. Justify this statements with the questions that illustrate the components of e-business models that can be developed as part of the strategic business/*IT* planning process.	08
		OR	
9	a	Explain how prototyping can be used as an effective technique to improve the process of system development for end users and IS specialists.	08
	b	Describe and give examples to illustrate how you might use each of the steps of the information systems development life cycle to develop and implement a business information system.	08
10	a	Identify and discuss the implementations process activities and timetables for a company installing an intranet-based employee benefits system in its human resource management department.	08
	b	Identify and discuss the implementation activities needed to transform a newly developed information system into an operational system for end users in the implementation process.	08
		OR	
11	a	Briefly discuss the different dimensions of ethical responsibility of business professionals.	08
	b	*IT* managers are increasingly concerned about securing the user end of their network and how they are facing this challenge? Discuss the security management tools and techniques for their networks.	08